

A STUDY OF THE PHYSICS AND CHEMISTRY OF L134N

J. E. DICKENS,¹ W. M. IRVINE, R. L. SNELL, E. A. BERGIN,² F. P. SCHLOERB, P. PRATAP,³ AND M. P. MIRALLES²
 Five College Radio Astronomy Observatory, 619 Lederle Graduate Research Center, University of Massachusetts, Amherst, MA 01003

Received 2000 February 1; accepted 2000 May 19

ABSTRACT

We have carried out a comprehensive and self-consistent study of the physical and chemical state of the core of the dark cloud L134N (L183), whose molecular abundances provide a standard against which chemical models may be compared. We used observations of the $\text{NH}_3(1, 1)$ and $(2, 2)$ rotation-inversion transitions to estimate the kinetic temperature, which was found to be consistent with 10 K and not varying with position. Densities were determined from multitransition statistical equilibrium calculations for HC_3N , N_2H^+ , and CS. The average density toward all lines of sight was $2 \times 10^4 \text{ cm}^{-3}$. As found by previous studies, the emission from various molecular species peaks in different positions: SO and SO_2 peak west of the central position, which is the location of the strongest emission from (e.g.) N_2H^+ and CH_3OH , with a second peak occurring for NH_3 and HC_3N to the north of the center. The most striking abundance variations occur in a north-south cut through the core center for HC_3N , C_2H , CS, SO, and SO_2 . A north to south decrease in the abundance of HC_3N and CS and a dramatic change in the CS/SO ratio, which has been shown to be a sensitive tracer of chemical evolutionary state, suggests that the north is at a younger evolutionary state than the south. Despite the “youth” of the N position, the CS/SO ratio suggests that it is still as “old” as or older than the most evolved region in TMC-1 (the northwest end of the ridge).

Subject headings: ISM: abundances — ISM: individual (L134N) — ISM: molecules

1. INTRODUCTION

Cold dark clouds have become important testing grounds for chemical models of the dense interstellar medium. The main reason for this is that the quiescent clouds lack embedded sources, whose energy input to the cloud can complicate the analysis by permitting a wider range of gas-phase reactions to occur and can cause the release of molecules formed on grain surfaces back into the gas phase. Despite the simpler physical properties of quiescent clouds, they do exhibit a complex inventory and distribution of chemical constituents. Therefore, they make a good laboratory for investigating the formation of molecules in gas-phase chemical models. This knowledge can then be extended to more complex regions, such as star-forming cores and giant molecular clouds associated with high-mass star formation. Furthermore, dark cloud cores are potential sites for future formation of solar type stars. Hence, they provide the initial boundary conditions for the chemical abundances and physical conditions within solar-like systems.

Accurate knowledge of the physical conditions within a molecular cloud is crucial to the interpretation of the observations and the extraction of chemical abundances. First, the excitation state of the molecule must be well understood in order to derive abundance information from molecular line data. Second, and perhaps more interestingly, the abundances of molecules are predicted to vary with physical conditions and with the evolutionary state of the cloud (e.g., Lee, Bettens, & Herbst 1996; Millar 1990; Graedel, Langer, & Frerking 1982). Comprehensive studies of the coupled relationship between physical conditions and chemical

abundances requires observations of emission from tracers of physical conditions and from chemically interesting species over a wide range of positions within a cloud. Such studies were done previously by our group for the giant molecular clouds Orion, M17, and Cepheus A (Ungerechts et al. 1997; Bergin et al. 1997), and more recently for the dark cloud TMC-1 (Pratap et al. 1997).

Here, we present the results of our comprehensive study of the cold dark cloud core L134N (also known as L183). This cloud core is located at a high Galactic latitude of 36° , and it is isolated and uncluttered by other emission along the line of sight. At a distance of 160 pc (Snell 1981), it is still well within the galactic disk and can be observed at high spatial resolution. We chose L134N in order to investigate differences between it and physically similar clouds such as the well-studied object TMC-1. Although apparently similar in kinetic temperature and density to TMC-1, possible chemical differences between L134N and TMC-1 have been noted by several authors. Specifically, L134N is observed to be more abundant in oxygen-rich molecules and highly reduced species, such as NH_3 and H_2S , while TMC-1 is more abundant in carbon-rich chains (Irvine 1992 and references therein; Ohishi, Irvine, & Kaifu 1992). Gradients in chemical abundances within each of these dark clouds have also been studied, most recently by Pratap et al. (1997) and Hirahara et al. (1995) for TMC-1 and by Swade (1989b) for L134N. The results from both studies of TMC-1 suggested a gradient in the chemical evolutionary state along the dense ridge, possibly resulting from a small gradient in the density. Swade (1989b) suggested a variation in the C/O abundance ratio, caused by depletion of water onto grains in the densest regions, to explain the chemical abundance variations seen in L134N. However, Swade (1989b) assumed that the emission was optically thin and that the energy levels of a given molecule were populated according to a Boltzmann distribution at a single excitation temperature. Pratap et al. (1997) found for TMC-1 that even some of the rarer isotopomer transitions were optically

¹ Present Address: MS 169-506, Jet Propulsion Laboratory, 4800 Oak Grove Drive, Pasadena, CA 91109; James.Dickens@jpl.nasa.gov.

² Harvard-Smithsonian Center for Astrophysics, 60 Garden Street, Cambridge, MA 02138.

³ MIT Haystack Observatory, Route 40, Westford, MA 01886.

TABLE 1
TRANSITIONS OBSERVED IN L134N

Species	Transition	ν (GHz)	η_B
CO	$J = 1 \rightarrow 0$	115.271203	0.41
^{13}CO	$J = 1 \rightarrow 0$	110.201370	0.46
C^{18}O	$J = 1 \rightarrow 0$	109.782182	0.46
CS	$J = 1 \rightarrow 0^{\text{a,b}}$	48.991008	0.45
	$J = 2 \rightarrow 1^{\text{a}}$	97.981040	0.50
C^{34}S	$J = 2 \rightarrow 1^{\text{a}}$	96.412940	0.51
HCO^+	$J = 1 \rightarrow 0$	89.188523	0.54
H^{13}CO^+	$J = 1 \rightarrow 0$	86.754329	0.55
N_2H^+	$J = 1 \rightarrow 0$	93.173830	0.51
CN	$N = 1 \rightarrow 0, J = 3/2 \rightarrow 1/2$		0.43
	$F = 3/2 \rightarrow 1/2$	113.488140	
	$F = 5/2 \rightarrow 3/2$	113.490982	
C_2H	$N = 1 \rightarrow 0, J = 1/2 \rightarrow 1/2$		0.55
	$F = 1 \rightarrow 1$	87.402004	
	$F = 0 \rightarrow 1$	87.407165	
SO	$J_K = 3_2 \rightarrow 2_1$	99.299875	0.49
	$J_K = 4_5 \rightarrow 4_4$	100.029627	0.49
	$J_K = 2_3 \rightarrow 1_2$	109.252134	0.46
^{34}SO	$J_K = 3_2 \rightarrow 2_1$	97.715390	0.50
SO_2	$3_{1,3} \rightarrow 2_{0,2}$	104.029420	0.48
HCN	$J = 1 \rightarrow 0$	88.631847	0.54
H^{13}CN	$J = 1 \rightarrow 0$	86.340184	0.56
HNC	$J = 1 \rightarrow 0$	90.663593	0.52
HN^{13}C	$J = 1 \rightarrow 0$	87.090851	0.55
CH_3OH	$J_K = 2_K \rightarrow 1_K$ quartet:		0.51
	$2_{-1} \rightarrow 1_{-1}\text{E}$	96.739390	
	$2_0 \rightarrow 1_0\text{A}^+$	96.741420	
	$2_0 \rightarrow 1_0\text{E}$	96.744580	
	$2_1 \rightarrow 1_1\text{E}$	96.755510	
HC_3N	$J = 4 \rightarrow 3^{\text{b}}$	36.369224	0.57
	$J = 10 \rightarrow 9$	90.979023	0.53
	$J = 12 \rightarrow 11$	109.173634	0.46
CH_3CCH	$J = 6 \rightarrow 5:$		0.48
	$K = 0$	102.547983	
	$K = 1$	102.546023	
	$K = 2$	102.540143	
NH_3	$(J, K) = (1, 1)^{\text{b}}$	23.694495	0.43
	$(J, K) = (2, 2)^{\text{b}}$	23.722633	0.43

NOTES.—Unless otherwise noted, all data were taken at FCRAO and all frequencies are the same as those used by Pratap et al. 1997, who list references.

^a CS and C^{34}S frequencies are taken from Kuiper et al. 1996.

^b NH_3 , HC_3N $J = 4 \rightarrow 3$, and CS $J = 1 \rightarrow 0$ data were taken at Haystack Observatory.

thick for physical conditions very similar to those in L134N. A new, more comprehensive study of L134N thus seemed useful.

We observed 34 transitions from 14 molecular species and their isotopomers at 120 positions in L134N. Statistical equilibrium calculations were performed to determine the kinetic temperature (T_K) from NH_3 observations and the molecular hydrogen density (n_{H_2}) from measurements of HC_3N , N_2H^+ , and CS emission (§ 3). Using these calculated temperatures and total H_2 densities, column densities and abundances relative to HCO^+ and H_2 were determined for all observed species in § 4. We compare our results with recent chemical models (§ 5) and conclude (§ 6) that the abundances and abundance gradients are consistent with what is expected for an evolving molecular cloud. Given the uniquely high abundances of molecules such as the carbon chains in TMC-1, L134N is probably a more typical dark

cloud than is TMC-1 (Caselli et al. 1998; Terzieva & Herbst 1998).

2. OBSERVATIONS

Observations were carried out between 1994 April and 1995 October utilizing the FCRAO⁴ 14 m antenna equipped with the QUARRY 3×5 beam focal plane array of receivers. An area of $\sim 8.5 \times 10'$ (R.A. $\times$ decl.; hereafter denoted α and δ) was mapped at 120 locations with spacings of $50''$ (equal to the HPBW at 100 GHz). The map center [$\alpha(1950) = 15^{\text{h}}51^{\text{m}}30^{\text{s}}$, $\delta(1950) = -2^{\circ}43'31''$] is that used by Swade (1989a), such that the results of both studies can be compared easily. The data were obtained by position switching to a fixed reference position [$\alpha(1950) = 15^{\text{h}}50^{\text{m}}46^{\text{s}}$, $\delta(1950) = -2^{\circ}32'31''$], which was observed to be free of ^{12}CO emission. The back end consisted of autocorrelation spectrometers with 1024 channels and 20 MHz bandwidth each, resulting in spectral resolution of $\sim 0.07 \text{ km s}^{-1}$ at 100 GHz.

The standard chopper wheel calibration was used, placing the observations on the T_A^* scale. The main beam efficiency (η_{mb}), which corrects for the coupling of the source to the antenna under the assumption that the source is uniform and fills only the main beam of the antenna (Kutner & Ulich 1981), was estimated from measurements of the planets made every 5 GHz between 90 and 115 GHz. Beam sizes ranged from $45''$ to $59''$. Table 1 contains a list of the line frequencies and main beam efficiencies for the molecular transitions observed.

Follow-up observations of the C^{34}S $J = 2 \rightarrow 1$ and H^{13}CO^+ $J = 1 \rightarrow 0$ transitions were made at the FCRAO 14 m antenna in 1998 May with the SEQUOIA 4×4 beam focal plane array of receivers. Since the QUARRY and SEQUOIA arrays have slightly different spatial configurations, the new maps were made with $25''$ spacings and resampled onto the QUARRY grid. The data were obtained by frequency switching ± 4 MHz. The spectrometer setup and beam efficiencies are the same as described previously.

In addition to the 3 mm mapping at FCRAO, observations of the HC_3N $J = 4 \rightarrow 3$, $\text{NH}_3(1, 1)$ and $(2, 2)$, and CS $J = 1 \rightarrow 0$ transitions were observed at the Haystack Observatory⁵ 37 m antenna during several sessions between 1995 May and 1997 July. The data were obtained by frequency switching 4.45 MHz of the 17.8 MHz total bandwidth. The velocity resolution at 36 GHz was $\sim 0.04 \text{ km s}^{-1}$. We calculated beam efficiencies by taking total power azimuth and elevation scans of Jupiter, Mars, and Venus and fitting for temperature and effective beam size. We adopted brightness temperatures at 36 GHz of 152, 194, and 466 K for Jupiter, Mars, and Venus, respectively (Greve et al. 1994). The calculated beam efficiencies are listed in Table 1, and the beam sizes were observed to be $96''$, $62''$, and $50''$ at 22, 36, and 49 GHz, respectively.

3. PHYSICAL CONDITIONS

3.1. Emission Characteristics

Figure 1 shows integrated intensity maps of the indicated transitions from key molecular species observed in our

⁴ The Five College Radio Astronomy Observatory is operated with the permission of the Metropolitan District Commission, Commonwealth of Massachusetts and with support from the National Science Foundation.

⁵ Radio Astronomy at the Haystack Observatory of the Northeast Radio Observatory Corporation (NEROC) is supported by a grant from the National Science Foundation.

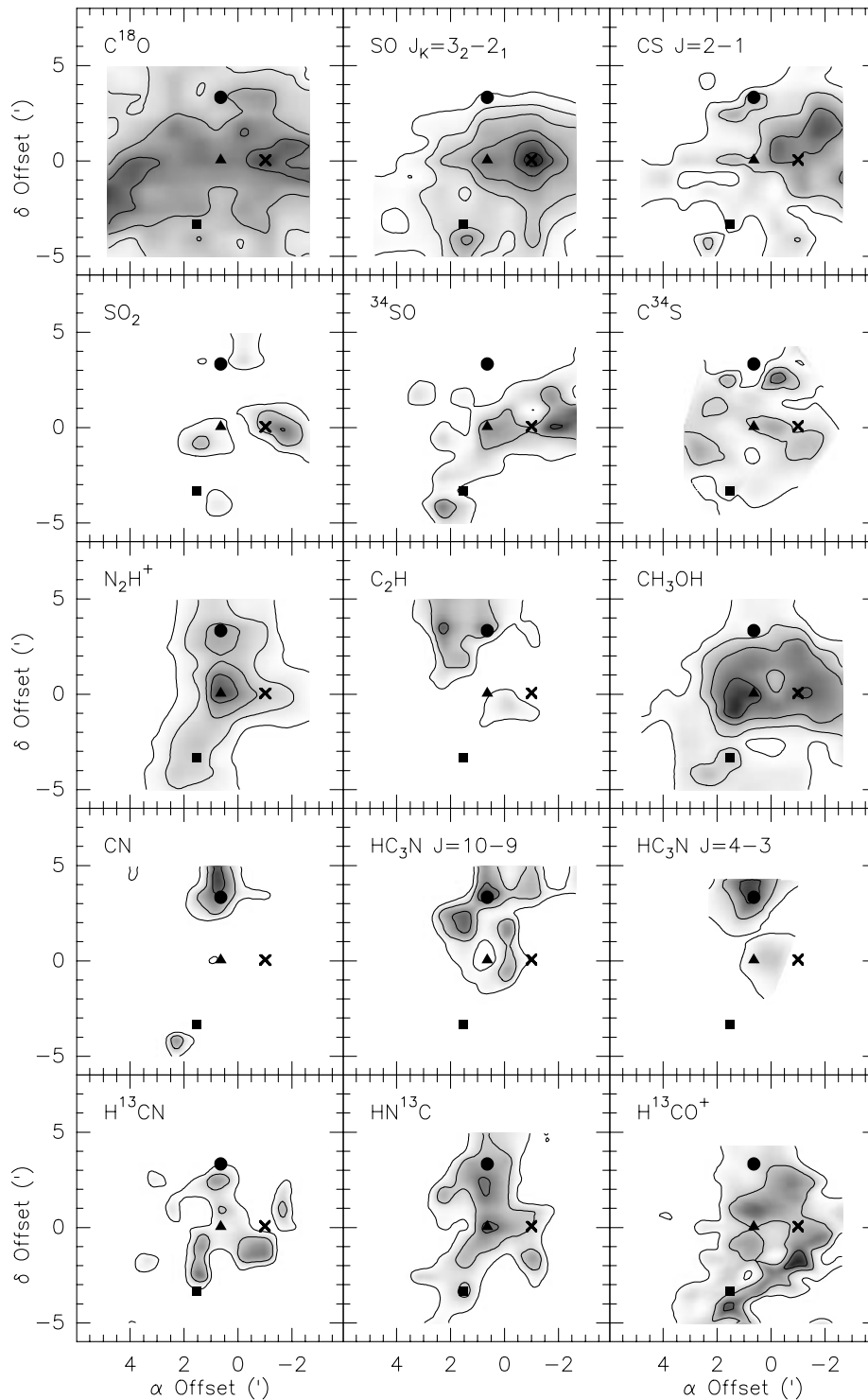


FIG. 1.—Integrated intensity maps for selected molecular transitions toward L134N. The (0, 0) position is ($15^{\text{h}}51^{\text{m}}30^{\text{s}}$, $-2^{\circ}43'31''$). The filled square marks the south (S) position ($+1.53$, -3.33), the filled triangle marks the center (C) position ($+0.64$, $+0.04$), the filled circle marks the north (N) position ($+0.64$, $+3.33$), and the cross marks the west (W) position (-1.01 , $+0.04$), as described in the text (§ 3). The maps for N_2H^+ , C_2H , CN , HC_3N , H^{13}CN , and HN^{13}C include all hyperfine components which were within the observing band. The CH_3OH map includes emission from both the A and E species. See list of transitions in Table 1.

survey. The most striking differences in the emission patterns can be seen when comparing the C^{18}O , SO , N_2H^+ , and CH_3OH maps. The C^{18}O emission is widespread and shows only a slight local peak west of the central (0, 0) position. Toward that same westward location, we see a maximum in the SO emission, and we have named this the

“west (W)” position (-1.01 , $+0.04$) and denoted it with a cross on the maps. This is our closest position to the SO peak discussed in Swade (1989a). We also note a local maximum in the CH_3OH and SO_2 emission within one beam of this position. While N_2H^+ has extended emission toward the W position, its distribution has a more pro-

nounced north-south trend, which can also be seen in HN^{13}C and to some extent in HC_3N . Additionally, we have marked three other positions as significant in our study, and we will continue to refer to them throughout the manuscript. The “north (N)” position ($+0.64, +3.33$) is marked with a filled circle and was chosen based on the observed local emission peaks in N_2H^+ , NH_3 (not included in Fig. 1), HN^{13}C , CN , and HC_3N . Using similar criteria, we defined a “center (C)” position ($+0.64, +0.04$), which is the location of the strongest N_2H^+ emission within our region, and labeled it with a filled triangle. The “south (S)” position ($+1.53, -3.33$) is denoted by a filled square and was chosen mainly from the N_2H^+ and HN^{13}C maps. However, there also seem to be local peaks in SO , SO_2 , CH_3OH , and HCO^+ . The integrated intensity maps thus reveal two basic morphologies: a north-south pattern represented by the carbon and nitrogen species (N_2H^+ , HCN , HNC , HC_3N , and to some extent CN and C_2H), and a lobate pattern centered around the C and W positions represented by the sulfur and oxygen species (SO , SO_2 , CH_3OH).

Determining the connection between the physical conditions, such as density and temperature, and the chemistry from integrated intensity maps is complicated by several factors. There may be density, temperature, and abundance gradients within any single beam or along any line of sight. Our observations represent the beam-averaged emission. High optical depths of some transitions will obscure the emission from deep within the cloud, making our task of untangling the above effects even more difficult. Our first step in analyzing this data set is thus to determine whether the emission from these molecules comes from the same volume of gas (§ 3.2). We find that this will greatly simplify our analysis, since we will not need to determine density, temperature, and column density for different velocity components. With knowledge of the velocity structure, we then assign a temperature (§ 3.3) and density (§ 3.4) along each line of sight.

3.2. Velocity Structure

Both our maps (Fig. 1) and those of L134N by Swade (1989a) reveal complex molecular emission distributions, which are suggestive of real chemical abundance and density gradients. With such variations in these two-dimensional maps, it would not be surprising to find evidence for variations along a single line of sight as well, perhaps the result of fragmentation of the molecular cloud into cores which might emit at different velocities. In order to investigate velocity structure in our data set, we examine velocity channel maps and line profiles of the various molecular transitions along selected lines of sight. We note that for the molecules exhibiting the strongest and most widespread emission, such as C^{18}O , N_2H^+ , and CH_3OH , the average line width toward all lines of sight is $\sim 0.47 \text{ km s}^{-1}$. The HN^{13}C is broadened by hyperfine components separated by 20 kHz ($\sim 0.07 \text{ km s}^{-1}$).

Figure 2 shows velocity maps of the integrated intensity from 2.2–3.0 km s^{-1} in steps of 0.2 km s^{-1} for C^{18}O , N_2H^+ (integrated over all hyperfine components in the observing band), CH_3OH (including both A and E species), SO ($J_K = 3_2 \rightarrow 2_1$), and CS ($J = 2 \rightarrow 1$). For the most part, we see that the emission from C^{18}O , N_2H^+ , and CH_3OH is similar in morphology to their total integrated intensity maps shown in Figure 1. Furthermore, the bulk of the emission occurs within the 2.4–2.6 and 2.6–2.8 km s^{-1} bins. From these

observations, we conclude that the emission from these molecules most likely originates from a single source. The C^{18}O emission is more extended toward lower velocities, but this is most likely the result of its lower excitation requirements, such that it is more sensitive to extended gas along the line of sight. The SO velocity maps, on the other hand, show a more complex velocity structure. The emission is widespread and fairly uniform between 2.2 and 2.6 km s^{-1} , but the emission becomes concentrated toward the W position at velocities $\gtrsim 2.8 \text{ km s}^{-1}$. The CS velocity maps show a similar extension toward this position at velocities $\gtrsim 2.8 \text{ km s}^{-1}$.

In order to determine whether we can associate this extended emission with multiple velocity features along particular lines of sight, we have plotted the spectra observed toward the N and W (Fig. 3), and C and S positions (Fig. 4). The spectra shown in these figures are spatially averaged over a 1.5 radius around each position to improve the signal-to-noise ratio. Most of the spectra are Gaussian shaped and centered near 2.5 km s^{-1} . However, the SO and C^{18}O spectra toward the W and S positions show convincing evidence for emission enhancement at velocities $\gtrsim 3 \text{ km s}^{-1}$.

Toward the C and the W positions (Fig. 5) we see a double-peaked line emission profile in the $\text{CS } J = 2 \rightarrow 1$ spectrum, while the $\text{CS } J = 1 \rightarrow 0$ and $\text{C}^{34}\text{S } J = 2 \rightarrow 1$ profiles are single-peaked. We have taken care to use the best determined frequencies for all three CS transitions (Kuiper, Langer, & Velusamy 1996). The C^{34}S transition lines up best with the 2.5 km s^{-1} velocity peak of CS and not with the dip, as would be expected for self-absorption. In addition to having a second peak at 3.0 km s^{-1} , the $\text{CS } J = 2 \rightarrow 1$ spectrum also appears to have a bump of emission near 3.4 km s^{-1} . Although the $J = 1 \rightarrow 0$ spectrum is single-peaked at 2.5 km s^{-1} , it is asymmetric with an extended shoulder of emission at the higher velocities. We interpret these observations to mean that there are two distinct velocity components centered at 2.5 and 3.0 km s^{-1} , but we do not have the signal-to-noise to see the 3.0 km s^{-1} component in C^{34}S . Furthermore, especially toward the W position, we see an extended tail of emission near 3.5 km s^{-1} , hinting at the possibility of a third velocity component.

We see further evidence for at least two velocity components in the HCO^+ spectra in Figure 6 for the C and W positions. The main isotopomer HCO^+ reveals extreme self-absorption at 2.5 km s^{-1} plus a second velocity component at 3.0 km s^{-1} which is not evident in the H^{13}CO^+ spectra. Langer et al. (1978) reported this same result close to our S position [at ($2', -4'$) relative to our central position]. The channel maps of Figure 7 verify that most of this higher velocity emission is located near the W position.

This velocity structure is also seen in the data of Swade (1989a). Indeed, while most molecular emission peaks at 2.5 km s^{-1} , Swade's spectra (Figs. 5–29) show weak emission wings and satellite peaks centered near 3.3 km s^{-1} for $\text{C}^{18}\text{O } J = 1 \rightarrow 0$ (e.g., Fig. 6b), $\text{CS } J = 2 \rightarrow 1$ (Fig. 7b), $\text{H}^{13}\text{CO}^+ J = 1 \rightarrow 0$ (Fig. 10b), and $\text{C}_3\text{H}_2 2_{1,2} \rightarrow 1_{0,1}$ (Fig. 16b). Similarly, Rydbeck et al. (1980) concluded that the width and shape of the $\text{SO } J_K = 1_0 \rightarrow 0_1$ transition toward the central position suggested the presence of at least two velocity components.

In summary, there is evidence for at least two velocity components and possibly a third in L134N. The main com-

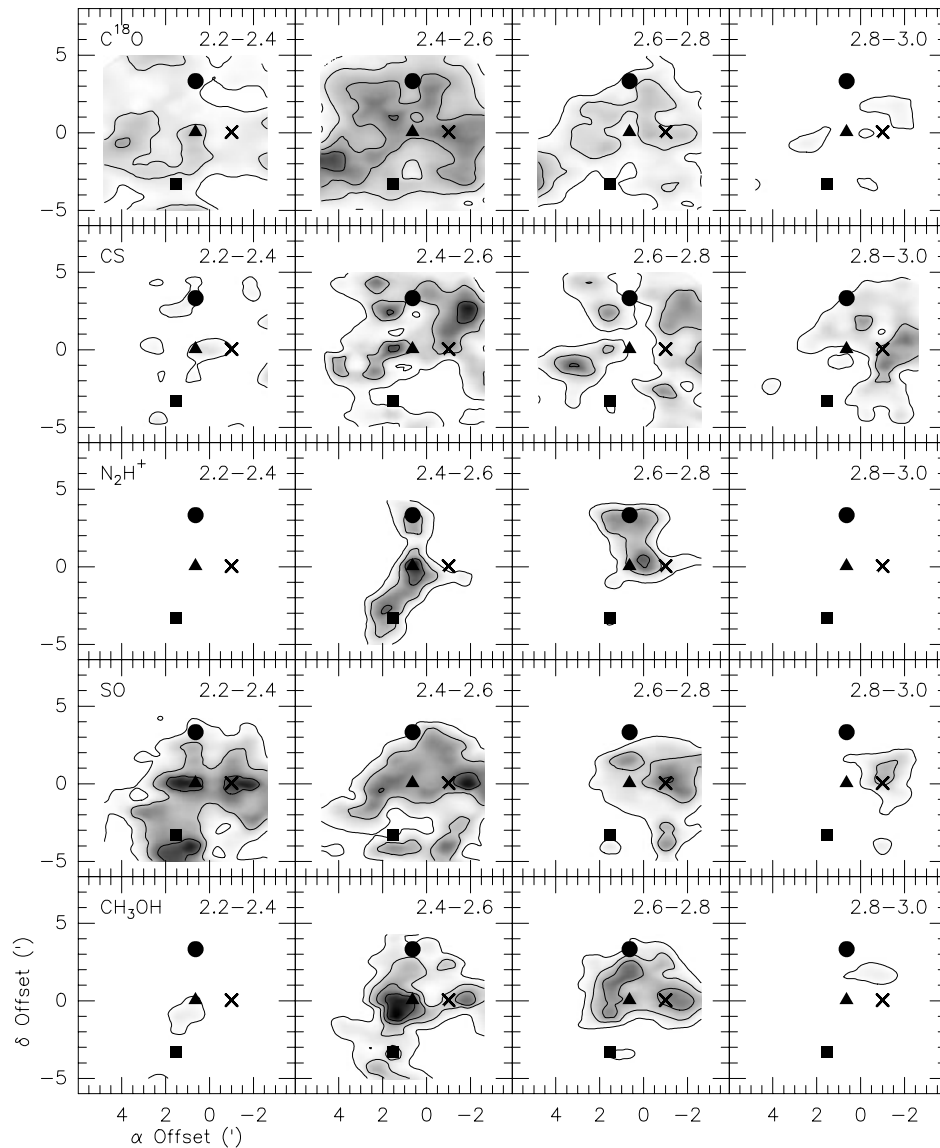


FIG. 2.—Velocity maps of the C^{18}O (1–0), CS (2–1), N_2H^+ (1–0), SO ($J_K = 3_2-2_1$), and CH_3OH (2_0-1_0 A⁺) emission in L134N from 2.2–3.0 km s^{-1} , shown in 0.2 km s^{-1} intervals. The (0, 0) position and the positions marked by symbols are the same as in Fig. 1.

ponent is centered at 2.5 km s^{-1} and is prevalent throughout the cloud in most molecular species. HCO^+ is extremely self-absorbed at these velocities toward the C and W positions. The second velocity component near 3.0 km s^{-1} is mainly associated with the W position and can be seen in the spectra of the most abundant species, such as C^{18}O , CS, SO, and HCO^+ . There is also a hint of a third component in the CS (2–1) spectra toward the C and W positions, but it is not evident in the spectra of other transitions. Since the second and third velocity components are not seen in the optically thinner transitions, we have no handle on the optical depth or excitation of these clumps. Therefore, we concentrate solely on the 2.5 km s^{-1} emission feature for the rest of our analysis and save the other components for future investigations.

3.3. Temperature

As mentioned in § 1, one of the goals of this project is to compare our results with those of Pratap et al. (1997) for the dark cloud TMC-1, which requires being as consistent as

possible with their analysis. Symmetric top molecules, such as CH_3CCH and NH_3 , are good tracers of kinetic temperature, since radiative transitions obey the selection rule $\Delta K = 0$ (K is the component of the total angular momentum along the molecular axis of symmetry). Hence, the different K -ladders are connected only through collisions, and the total population of one K -ladder relative to another should exhibit a thermal distribution at the kinetic temperature of the gas along the line of sight. Pratap et al. (1997) relied primarily on observations of the $K = 0, 1$, and 2 ladders of the $\text{CH}_3\text{CCH } J = 6 \rightarrow 5$ transition to obtain estimates of the kinetic temperature in TMC-1. Unfortunately, we did not detect CH_3CCH toward any positions in L134N. However, Pratap et al. (1997) compared their CH_3CCH kinetic temperatures with corresponding values derived from observations of the $(J, K) = (1, 1)$ and $(2, 2)$ rotation-inversion transitions of NH_3 and found the two estimates to be consistent within the uncertainties of the observations. Hence, we relied on observations of these NH_3 rotation-inversion transitions toward 11 positions in

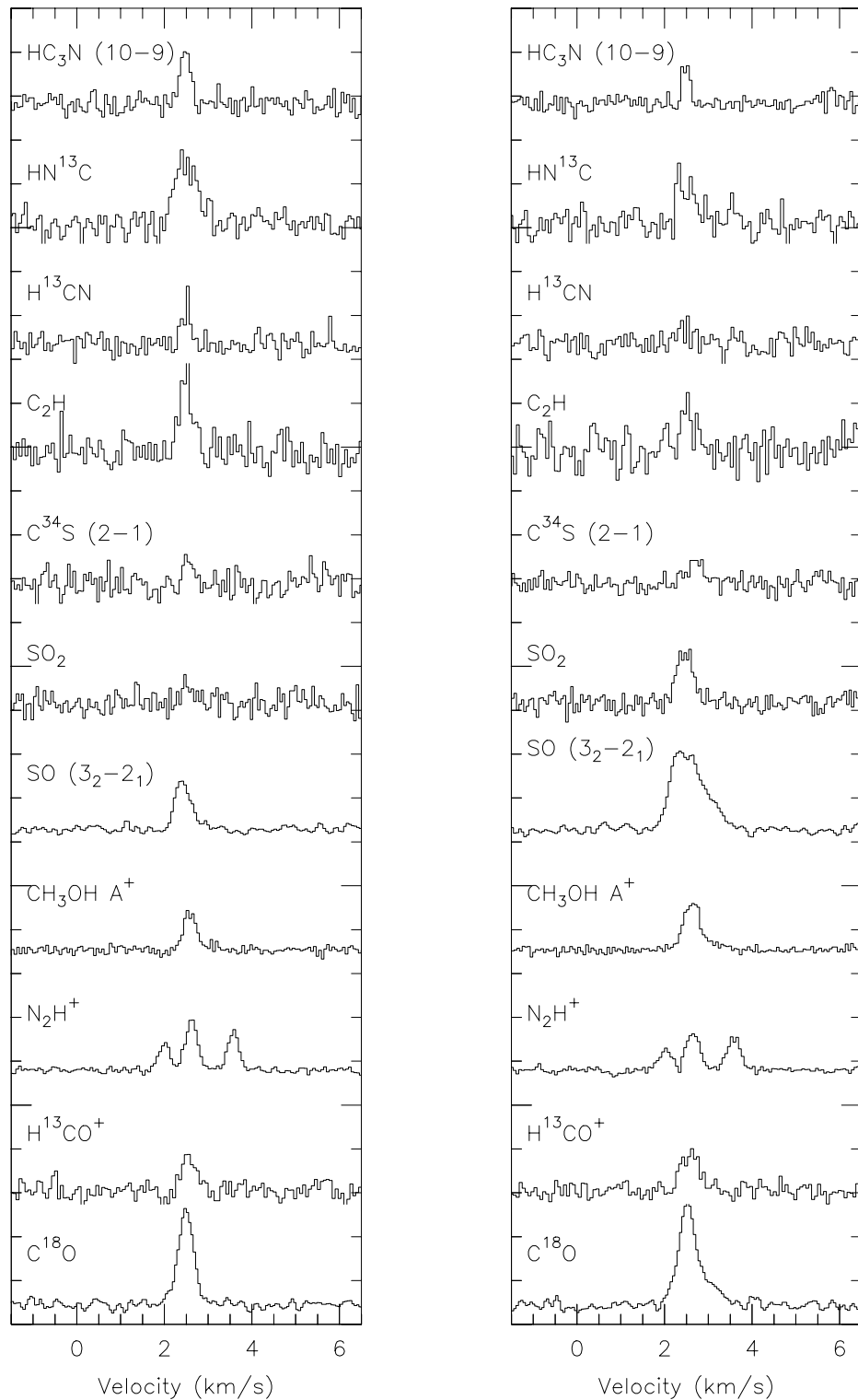


FIG. 3.—Spectra observed toward the N (left panel) and W (right panel) positions (as defined in Fig. 1 and § 3) for selected transitions in Table 1. To improve the signal-to-noise ratio, we have spatially averaged 1.5 radius regions around each position. The y-axis is presented in the T_A^* scale in units of (K). The bottom five spectra [$C^{18}O$, $H^{13}CO^+$, N_2H^+ , $CH_3OH A^+$, and $SO (3_2-2_1)$] all have the same y-scale, and the remaining spectra are magnified by a factor of 3. The scale remains fixed from position to position in order to facilitate direct comparison. For reference, the $C^{18}O$ spectrum toward the N position has a peak antenna temperature, $T_A^* = 1.5$ K.

L134N in order to obtain reliable estimates of the kinetic temperature. Only upper limits are reported for the (2, 2) line intensity at two of these positions.

Rotation temperatures are calculated using the technique described in Bachiller, Guilloteau, & Kahane (1987). The 18

hyperfine components of the (1, 1) transition were simultaneously fitted for velocity, line width, optical depth of the main group of blended hyperfine transitions, and the product $X = \tau \times (T_{ex} - T_{bg})$. As pointed out by Bachiller et al. (1987), this last quantity has the advantage that when the

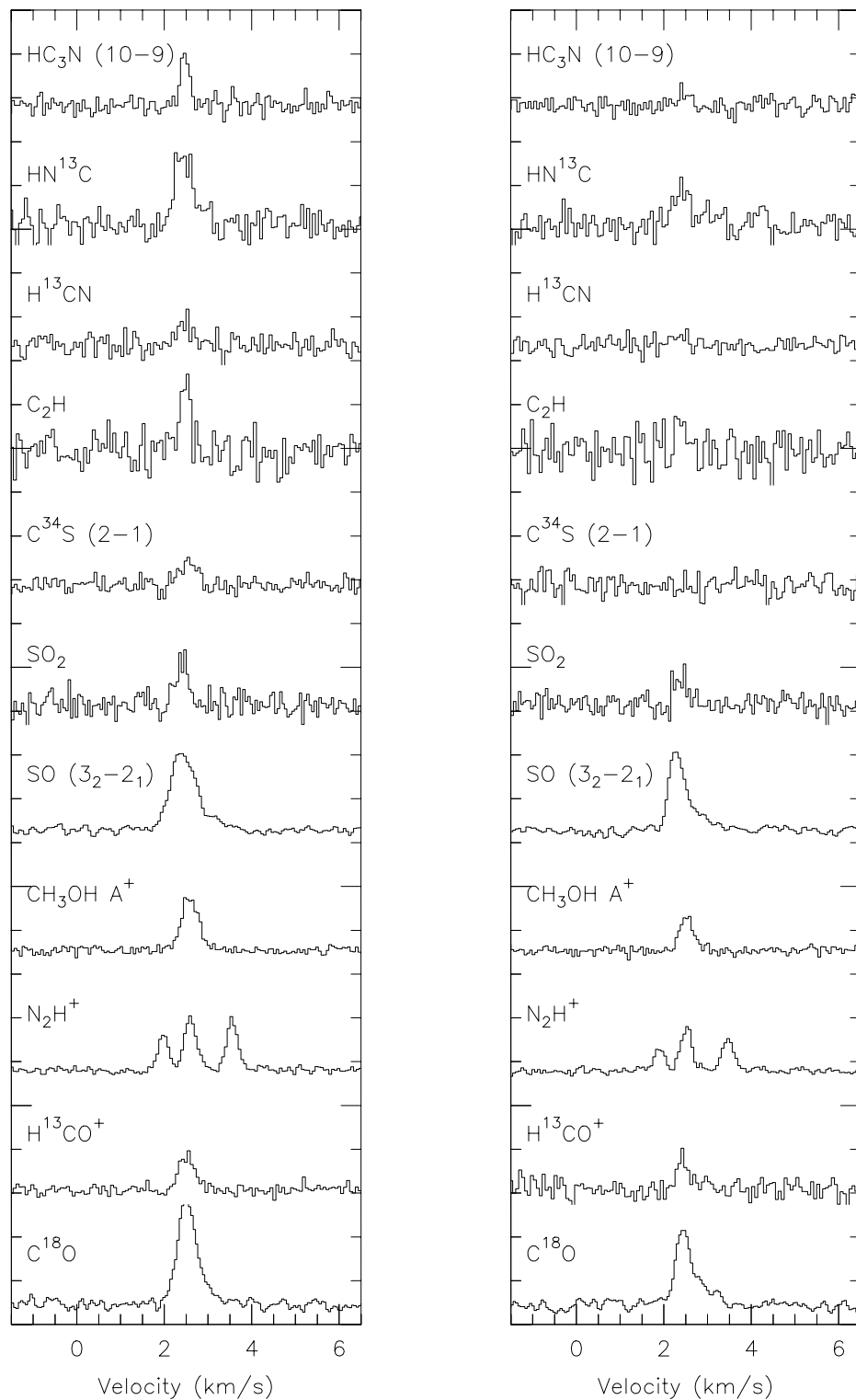


FIG. 4.—Spectra observed toward the C (left panel) and S (right panel) positions. The format is the same as in Fig. 3.

transition is optically thin, one can still obtain fit results even though the spectrum is no longer dependent on optical depth. In the case of the (2, 2) transition, the main group of hyperfine components is well blended and the satellite hyperfine components are too weak (this transition is 42 K above the ground state) to fit in the same manner as the (1, 1) transition. We merely fit a single Gaussian to the main group blend. Then, the rotational temperature is related to the ratio of the (1, 1) and (2, 2) main group integrated inten-

sities, both of which contain one-half the total transition line strength, by the expression given in Bachiller et al. (1987). This method assumes equal excitation among the two metastable levels. The uncertainty in the rotational temperature is determined by propagating the fitted parameter uncertainties.

To relate the rotational temperature to a kinetic temperature, the “three-level” statistical equilibrium model described by Walmsley & Ungerechts (1983) was adopted.

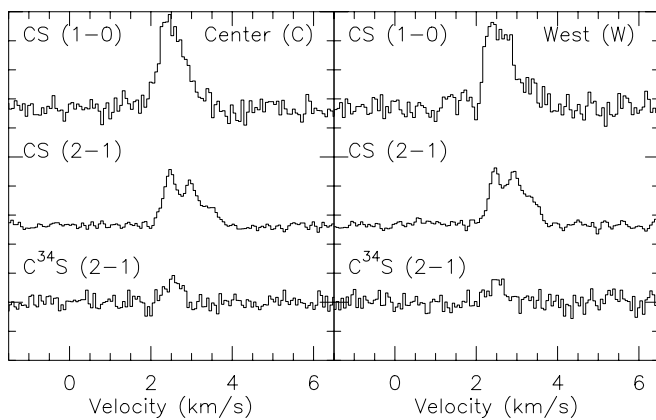


FIG. 5.—CS spectra toward the C (left panel) and W (right panel) positions. The y-axis is presented in the T_A^* scale in degrees Kelvin, and the y-scale is the same for each transition with 0.2 K between tick marks.

Since the (2, 2) level is 42 K above the ground state, it can be assumed that higher energy levels are unpopulated and that our three-level model is appropriate for temperatures less than 40 K. Applying detailed balance and the Boltzmann equation, Walmsley & Ungerechts (1983) derive the relationship between rotational temperature and kinetic

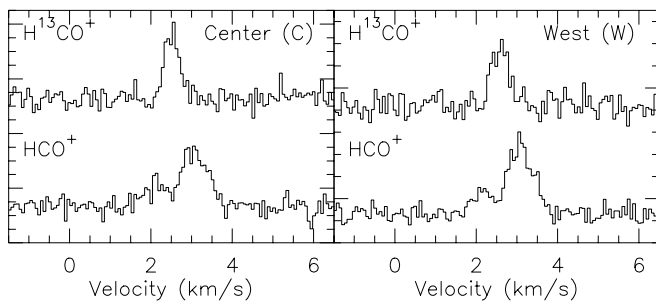


FIG. 6.— H^{13}CO^+ and HCO^+ spectra toward the C (left panel) and W (right panel) positions. The y-axis is presented in the T_A^* scale in degrees Kelvin, and the y-scale is the same for each transition with 0.2 K between tick marks.

temperature as equation (2) in their manuscript. Collision coefficients are taken from Danby et al. (1988). The kinetic temperature is assumed to have the same percentage uncertainty as the rotational temperature determined previously. Typically, the derived rotational and kinetic temperatures differ by only 0.2–0.3 K.

The rotational temperature and kinetic temperature results are shown in Table 2. The kinetic temperatures range from 9.1 ± 0.3 to 10.7 ± 1.0 K, which are consistent with a uniform kinetic temperature of 10 K throughout the cloud. All of our subsequent analyses assume that the kinetic temperature is 10 K and does not vary with position. Swade (1989b) quotes a uniform temperature of 12 K from his analysis of the NH_3 emission. Therefore, we examined the consequence of a ± 2 K change in temperature on the total densities and column densities and discuss this in § 4.

3.4. Density

As in the case of the TMC-1 study by Pratap et al. (1997), we had hoped to rely on multiple transition measurements of HC_3N to determine H_2 densities in L134N. Unfortunately, the emission from this molecule is weaker and less widely distributed in L134N than in TMC-1. We therefore relied on observations of CS and N_2H^+ to determine densities at many locations in the cloud. Using multiple transitions of CS and its isotopomer C^{34}S doubled the number of lines of sight for which we could find densities and hence abundances, and it gave us data in the southern region of the cloud. Additionally, N_2H^+ has widespread emission (cf. Fig. 1), a large dipole moment (cf. Table 3), and multiple hyperfine components, making it a sensitive probe of the densest gas along any line of sight. There were a sufficient number of lines of sight with densities determined from more than one of these probes to check that all three molecules were providing consistent results.

The determination of densities from multiple transitions was carried out using statistical equilibrium calculations, and the effects of radiative trapping were incorporated using the large velocity gradient (LVG) approximation for a spherical source. The parameters for the model are kinetic temperature, H_2 density, molecular species column density, and line width. The radiative and collisional rates for the

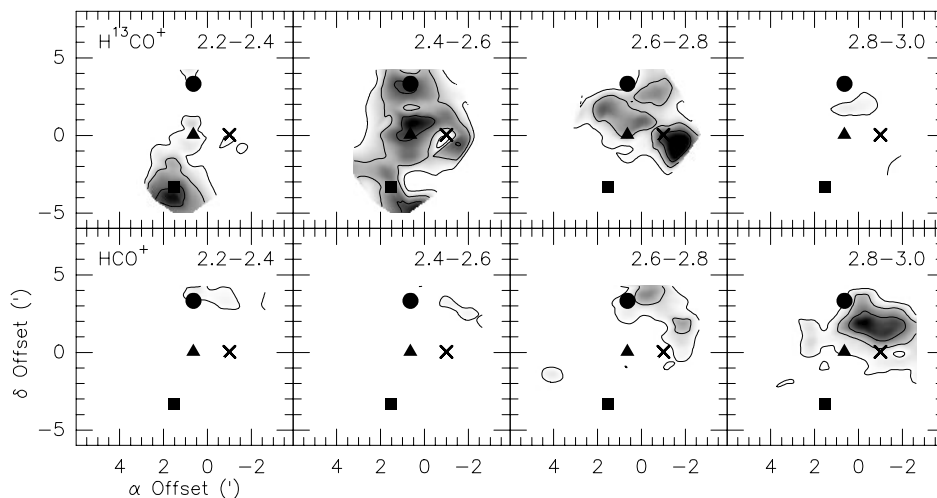


FIG. 7.— $\text{H}^{13}\text{CO}^+(1-0)$ and $\text{HCO}^+(1-0)$ velocity maps with the same format as Fig. 2

TABLE 2
KINETIC TEMPERATURES IN L134N FROM AMMONIA OBSERVATIONS

Offset α (arcmin)	Offset δ (arcmin)	X (K)	$\tau(1, 1)$	$\Delta v(1, 1)$ (km s ⁻¹)	$\int T_A^* \delta v(2, 2)$ (K km s ⁻¹)	T_{rot} (K)	T_K (K)
0.63	1.69	8.40(0.65)	3.5(0.4)	0.47(0.03)	0.142(0.019)	8.9(0.3)	9.1(0.3)
1.53	-1.69	8.17(0.60)	3.0(0.4)	0.33(0.02)	0.139(0.019)	9.7(0.4)	9.9(0.4)
1.53	0.04	8.88(0.49)	3.8(0.3)	0.36(0.01)	0.137(0.021)	9.3(0.3)	9.5(0.3)
-0.14	0.82	7.91(0.67)	3.8(0.5)	0.42(0.03)	0.147(0.021)	9.3(0.4)	9.6(0.4)
1.53	2.53	8.10(0.75)	3.6(0.6)	0.33(0.02)	0.112(0.022)	9.2(0.5)	9.4(0.5)
-1.88	-0.01	1.74(0.36)	1.5(0.5)	0.56(0.09)	0.069(0.019)	10.5(1.0)	10.8(1.0)
0.63	3.33	20.60(0.86)	7.4(0.5)	0.23(0.01)	0.209(0.043)	9.3(0.4)	9.6(0.5)
0.63	0.04	17.49(0.83)	6.3(0.5)	0.28(0.01)	0.182(0.025)	9.0(0.3)	9.2(0.3)
1.53	-2.50	9.91(0.57)	3.0(0.4)	0.25(0.01)	0.121(0.023)	9.6(0.5)	9.8(0.5)
3.17	0.04	1.08(0.27)	<0.3	0.49(0.10)	<0.213	<12.4	<12.8
-1.88	2.56	3.30(0.81)	<0.8	0.37(0.10)	<0.234	<10.2	<10.5

NOTES.—Map center (0, 0) corresponds to $\alpha(1950) = 15^{\text{h}}51^{\text{m}}30^{\text{s}}$, $\delta(1950) = -2^{\circ}43'31''$. $X = \tau(1, 1) \times (T_{\text{ex}} - T_{\text{bg}})$. Optical depth, $\tau(1, 1)$, obtained from hyperfine component intensity ratios, is that for the main group of blended hyperfine transitions, which contains one-half of the total transition intrinsic line strength. Uncertainties are 1σ and are included in parenthesis. Upper limits are 3σ .

transitions were taken from the literature (Table 3). A kinetic temperature of 10 K was assumed, based on the results in § 3.3, and the line widths were taken from the observations. The spectra were spatially averaged over a 1.5 radius to increase the signal-to-noise ratio. This was especially important for the determination of density from the relatively weak emission of HC₃N.

3.4.1. HC₃N Modeling

We observed the HC₃N $J = 4 \rightarrow 3$, $10 \rightarrow 9$, and $12 \rightarrow 11$ transitions but did not detect the $J = 12 \rightarrow 11$ transition toward any line of sight in L134N. The strongest emission from the other two lines was restricted to the northern region of the cloud (cf. Fig. 1). An important point is that these three transitions were observed with similar beam sizes, thus eliminating effects from different angular resolutions. The $J = 4 \rightarrow 3$ transition has the lowest critical density (10^4 cm^{-3}) and is therefore most sensitive to lower density gas. If L134N has a core-halo structure, then we might expect to see extended emission in the $J = 4 \rightarrow 3$ integrated intensity maps. This is not the case (Fig. 1). Furthermore, Gaussian fits to the $J = 4 \rightarrow 3$ and $10 \rightarrow 9$ lines reveal that they have similar line widths, centroid velocities, and

line shapes. Therefore, it is reasonable to assume that the emission originates from the same volume of gas. If a halo does exist, it does not have high enough density to excite the $J = 4 \rightarrow 3$ transition.

For our statistical equilibrium model, the collisional coefficients were taken for collisions between HC₃N and He (cf. Table 3). Since the more likely collision partner of HC₃N is H₂, we scale the collisional rates by the factor $2^{1/2}$ to account for the velocity difference between H₂ and He. Gaussian line profiles were fitted to the three resolvable hyperfine components of the $J = 4 \rightarrow 3$ transition and to the $J = 10 \rightarrow 9$ transition. Upper limits to the emission from the $J = 12 \rightarrow 11$ transition were determined by assuming that it has a Gaussian shape with line width equal to that found for the other transitions. These five measurements were used to constrain the remaining two free parameters, H₂ density and HC₃N column density, in our model. We have assumed a 15% calibration uncertainty added in quadrature sum with the intrinsic noise in the spectra. The relative strengths of the hyperfine components of the $J = 4 \rightarrow 3$ transition provide a constraint to the HC₃N optical depth, and the $J = 10 \rightarrow 9$ and $12 \rightarrow 11$ (upper limits) intensities provide a good measure of the excitation

TABLE 3
MOLECULAR DIPOLE MOMENTS AND COLLISION RATE COEFFICIENTS

Species	μ (D)	μ Reference	Collision Pairs	Collision Rate Reference
CO	0.11	Lovas & Krupenie 1974	CO-paraH ₂	Flower & Launay 1985
CS	1.96	Lovas & Krupenie 1974	CS-He	Green & Chapman 1978
HCN	2.99	Maki 1974	HCN-He	Green & Thaddeus 1974
HNC	2.99	Maki 1974	HCN-He	Green & Thaddeus 1974
CN	1.45	Thomson & Dalby 1968	CO-paraH ₂	Flower & Launay 1985
C ₂ H	0.80	Tucker et al. 1974	HCN-He	Green & Thaddeus 1974
N ₂ H ⁺	3.40	Green et al. 1974	N ₂ H ⁺ -He	Green 1975
HCO ⁺	3.97	Botschiwina 1989	N ₂ H ⁺ -He	Green 1975
HC ₃ N	3.72	Lafferty & Lovas 1978	HC ₃ N-He	Green & Chapman 1978
SO	1.55	Tiemann 1974	SO-H ₂	Green 1994
SO ₂	1.63	Lovas 1985	SO ₂ -He	Green 1995
CH ₃ CCH	0.78	Bauer et al. 1979	CH ₃ CCH-H ₂	Green 1986
CH ₃ OH	0.89	Lees 1973	CH ₃ OH-H ₂	Menten et al. 1988

NOTE.— μ is the dipole moment in debye.

conditions along the line of sight. This is the same technique as used for TMC-1 (Pratap et al. 1997).

For the 19 positions at which we had detections in both the $J = 4 \rightarrow 3$ and $10 \rightarrow 9$ (and upper limits on the $J = 12 \rightarrow 11$) transitions, we used a nonlinear least-squares program which varied the density and column density to minimize the reduced χ^2 to the observed intensities. Since the density and column density vary independently only in

the case of optically thin emission, we mapped χ^2 in two dimensions and found the maximum and minimum values of density and column density which produced a change in the minimum reduced χ^2 of 1 (see Bevington & Robinson 1992; Bergin, Snell, & Goldsmith 1996). The resultant densities are shown in Table 4. The optical depths derived using our model agree with those calculated from the relative strengths of the hyperfine components.

TABLE 4
RESULTS FROM N_2H^+ , CS, AND HC_3N DENSITY ANALYSIS FOR L134N

OFFSET α (arcmin)	OFFSET δ (arcmin)	log (n_{H_2})			log (n_{H_2}) Average	REDUCED χ^2
		N_2H^+	CS	HC_3N		
0.63	0.04	4.18(0.21)	4.01(0.28)	...	4.12(0.17)	0.86
0.63	0.88	4.12(0.32)	4.16(0.30)	...	4.14(0.22)	0.47
0.63	1.69	4.34(0.34)	3.99(0.21)	...	4.09(0.18)	1.98
0.63	2.53	4.40(0.23)	3.97(0.25)	...	4.20(0.17)	1.62
0.63	3.33	4.14(0.25)	4.66(0.24)	4.38(0.09)	4.38(0.08)	1.56
0.63	4.17	4.27(0.26)	...	4.33(0.07)	4.32(0.06)	0.96
0.63	-0.85	4.25(0.35)	4.21(0.29)	...	4.23(0.22)	0.89
0.63	-1.69	4.73(0.28)	4.41(0.23)	...	4.54(0.18)	1.03
0.63	-2.50	4.41(0.31)	...	...	4.41(0.31)	1.01
0.63	-4.98	4.40(0.36)	...	...	4.40(0.36)	0.80
1.53	0.04	...	4.02(0.21)	...	4.02(0.21)	1.11
1.53	0.88	...	...	4.74(0.28)	4.74(0.28)	0.17
1.53	1.69	...	...	4.73(0.10)	4.73(0.10)	0.15
1.53	2.53	4.31(0.33)	4.03(0.31)	...	4.16(0.22)	0.95
1.53	3.33	4.70(0.25)	...	4.27(0.13)	4.36(0.11)	0.75
1.53	4.17	4.39(0.26)	...	4.43(0.11)	4.42(0.10)	1.15
1.53	-0.85	4.44(0.35)	...	4.41(0.22)	4.42(0.19)	0.59
1.53	-1.69	...	...	3.76(0.23)	3.76(0.23)	2.34
1.53	-2.50	4.50(0.25)	3.97(0.38)	...	4.34(0.21)	1.11
1.53	-3.33	4.52(0.28)	...	...	4.52(0.28)	0.57
1.53	-4.14	...	4.30(0.24)	...	4.30(0.24)	0.60
1.53	-4.98	4.11(0.45)	...	...	4.11(0.45)	1.09
2.31	2.56	4.29(0.49)	...	4.78(0.09)	4.76(0.08)	0.75
2.31	3.46	4.37(0.37)	...	4.01(0.24)	4.12(0.20)	1.35
2.31	4.31	...	...	3.92(0.35)	3.92(0.35)	2.72
2.31	-0.01	...	4.02(0.26)	...	4.02(0.26)	0.44
2.31	-1.56	...	4.02(0.27)	...	4.02(0.27)	0.43
2.31	-2.46	5.04(0.43)	...	...	5.04(0.43)	2.01
2.31	-3.30	4.58(0.28)	...	...	4.58(0.28)	0.37
2.31	-4.20	4.75(0.45)	...	...	4.75(0.45)	1.63
3.17	0.04	...	4.01(0.26)	...	4.01(0.26)	0.48
3.17	-2.50	...	3.92(0.29)	...	3.92(0.29)	2.50
-0.14	0.82	4.34(0.25)	...	4.55(0.21)	4.46(0.16)	0.81
-0.14	1.72	4.31(0.31)	...	4.64(0.11)	4.61(0.10)	1.11
-0.14	2.56	...	...	4.11(0.10)	4.11(0.10)	1.44
-0.14	3.46	...	...	4.46(0.15)	4.46(0.15)	1.31
-0.14	4.31	...	...	4.66(0.15)	4.66(0.15)	0.75
-0.14	-0.02	4.64(0.36)	...	...	4.64(0.36)	1.01
-0.14	-0.72	5.06(0.34)	4.01(0.26)	...	4.41(0.21)	1.51
-0.14	-1.56	...	...	4.77(0.15)	4.77(0.15)	0.68
-1.01	0.04	4.18(0.34)	4.01(0.21)	...	4.06(0.18)	0.96
-1.01	0.88	...	4.01(0.25)	...	4.01(0.25)	1.64
-1.01	2.53	4.46(0.45)	...	...	4.46(0.45)	0.49
-1.01	3.33	4.15(0.33)	...	4.73(0.21)	4.56(0.18)	1.64
-1.01	4.17	...	...	4.82(0.11)	4.82(0.11)	0.68
-1.01	-3.33	...	4.00(0.29)	...	4.00(0.29)	0.91
-1.88	1.73	...	4.02(0.24)	...	4.02(0.24)	0.96
-1.88	2.56	...	4.03(0.24)	...	4.03(0.24)	1.37
-1.88	-0.01	...	4.00(0.29)	...	4.00(0.29)	1.25
-2.66	0.04	...	4.03(0.30)	...	4.03(0.30)	1.19

NOTES.—Offsets are from $\alpha(1950) = 15^h51^m30^s$, $\delta(1950) = -2^\circ43'31''$. Uncertainties are 1σ and are included in parenthesis.

3.4.2. N_2H^+ Modeling

We also model densities from the emission of the hyperfine components of the $J = 1 \rightarrow 0$ transition of N_2H^+ . We simultaneously fitted Gaussians to all seven hyperfine transitions with the constraint that they all share the same line width. The collisional coefficients were calculated from collisions with He (cf. Table 3), so we scaled the rates by $2^{1/2}$ to correct for the velocity difference between H_2 and He. The statistical equilibrium model was run as described previously in § 3.4.1 to minimize χ^2 . Our model assumes a filling factor of unity so that we can determine the excitation, and hence the density, directly from the optical depth and antenna temperature.

We detected emission from N_2H^+ toward 64 lines of sight in L134N. Since we did not have observations of higher energy transitions with which to constrain the excitation, the uncertainty in the density is larger than that for HC_3N . Furthermore, when the ratio of the hyperfine transitions is consistent with optically thin emission, we get no information on the optical depth of this transition, and our model finds solutions over a large range of densities. We only quote densities for those lines of sight where the uncertainties in the density are less than 0.5 in the log. This left us with 29 positions in L134N. The results are shown in Table 4. There are 10 positions with which we also have density estimates from HC_3N . The average N_2H^+ density for these overlapping points is $2.2 \times 10^4 \text{ cm}^{-3}$, consistent with that for HC_3N which is $2.8 \times 10^4 \text{ cm}^{-3}$. Individual points have derived values which agree to within a factor of 3 for these two probes.

3.4.3. CS Modeling

In order to gain density information in the southern region of L134N, we relied on observations of the $J = 1 \rightarrow 0$ and $2 \rightarrow 1$ transitions of CS, supplemented by observations of the $J = 2 \rightarrow 1$ transition of the $C^{34}S$ isotopomer. We had detections of all three transitions toward 23 lines of sight in L134N. As described in § 3.2, the CS $J = 2 \rightarrow 1$ transition contains evidence for multiple velocity components. Therefore, we fitted two Gaussian line profiles to this transition, with the line width of the 2.5 km s^{-1} component fixed at the value determined from the Gaussian fit to the $C^{34}S$ line, which did not have multiple velocity components (cf. Fig. 5). In the following, we use only the 2.5 km s^{-1} component of the $J = 2 \rightarrow 1$ line. We assumed that the CS $J = 1 \rightarrow 0$ emission was also due to a single velocity component. For most lines of sight, the $J = 1 \rightarrow 0$ transition had a larger line width than the $J = 2 \rightarrow 1$. This is most likely due to the ease with which this low energy (3 K) transition can be excited. We took the average of the $J = 2 \rightarrow 1$ and $1 \rightarrow 0$ line widths to put into our statistical equilibrium model.

We simultaneously fitted the emission from the three transitions, assuming $^{32}S/^{34}S = 14$ to determine the optical depth (see below) and varied the density and column density until we minimized χ^2 . The uncertainty in the density was estimated in the same manner as that described in § 3.4.1. Our model deduced optical depths much greater than 1 for both of the CS transitions. Therefore, the $J = 2 \rightarrow 1$ and $1 \rightarrow 0$ ratio is fairly insensitive to the density. Most of the density information comes from the excitation temperature, which was derived from the optical depth and antenna temperature of the CS and $C^{34}S$ $J = 2 \rightarrow 1$ transitions. Therefore, as was the case for the N_2H^+ model, this model assumes a filling factor of unity. We also tried a model

assuming $^{32}S/^{34}S = 22$, the terrestrial ratio, but the fits were poor, as was the case for TMC-1 (Pratap et al. 1997). An isotopic ratio of 14 is consistent with the determination of this ratio from Galactic cosmic rays (Thayer 1996), but this method has large uncertainties. The astronomical determinations are mixed: from extensive observations of SO_2 in the Orion molecular cloud, Schloerb et al. (1983) find a ratio of 16, while the most extensive study of the sulfur isotope ratios in Galactic star-forming regions is consistent with the terrestrial value (Chin et al. 1996). The ratio in dark clouds remains uncertain. The density determinations for CS are shown in Table 4. For the 11 positions for which we also derived densities from N_2H^+ , the average density from CS is $1.3 \times 10^4 \text{ cm}^{-3}$, approximately half that of N_2H^+ , which is $2.4 \times 10^4 \text{ cm}^{-3}$.

3.4.4. Density Summary

We have determined densities toward 50 lines of sight in L134N using statistical equilibrium calculations with HC_3N , CS, and N_2H^+ as our density tracers. The average density for all lines of sight is 2.5×10^4 for both HC_3N and N_2H^+ and $1 \times 10^4 \text{ cm}^{-3}$ for CS. The only position which has densities determined for all three probes is the N position (+0.63, +3.33), where the highest and lowest estimates are a factor of 3 different, and the average is the same as that determined for HC_3N . The HC_3N density probe is the most accurate, with typical uncertainties in the density of a factor of 1.5. The lower density values found using CS are consistent with what was seen for TMC-1 (Pratap et al. 1997). This could imply that CS is more sensitive to less dense gas along the line of sight than are HC_3N and N_2H^+ . Alternatively, since these averages include points where there is no overlap between the three tracers, it could be that the gas is indeed less dense toward positions where only CS is detected.

Although our results, particularly when combined with the TMC-1 experience (Pratap et al. 1997), suggest that CS may yield slightly lower densities than HC_3N and N_2H^+ , we feel that using any one tracer in preference to another might be biased. Therefore, we calculate a single density as the average of all of the three tracer densities (weighted by the uncertainties) observed toward a given line of sight. The results are shown in Table 4. The reduced χ^2 shown in the last column is determined by refitting each of the observed transitions for the three molecules to a model which is constrained to have the average density. To see whether our model is an accurate representation of the data, we have examined the distribution of reduced χ^2 . For a normal distribution with 8 degrees of freedom, we expect a median reduced χ^2 of 0.9 and fewer than 5 positions with a value less than 0.44 or greater than 1.67 (Bevington & Robinson 1992). We have a median χ^2 of 0.96, with 5 points less than 0.44 and greater than 1.67. Therefore, we believe that modeling each line of sight as a single density is an adequate representation of the observed data.

The upper left panel of Figure 8 shows a north-center-west cut of the derived densities. There is little evidence for a density gradient in these plots, and the average density along all lines of sight is $2.1 \times 10^4 \text{ cm}^{-3}$. For comparison, Swade (1989b) quotes a density of $3 \times 10^4 \text{ cm}^{-3}$ from statistical equilibrium calculations of the $NH_3(1, 1)$ rotation-inversion emission. Madden (1990) finds the same result for multitransition observations of the cyclic molecule C_3H_2 . Snell, Langer, & Frerking (1982) found densities of 10^4

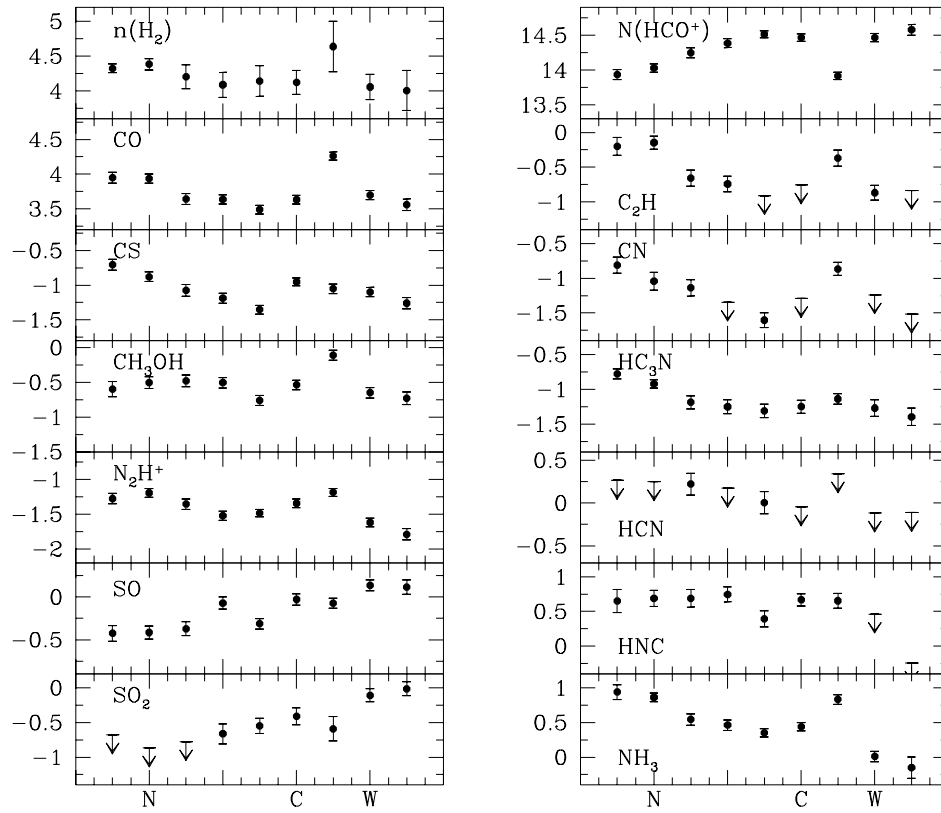


FIG. 8.—North-center-west cut through the cloud core showing log of the density in the upper left panel and the log of the HCO^+ column density in the upper right panel. The rest of the panels show the log of the abundance of the indicated molecule relative to HCO^+ (derived from H^{13}CO^+). The error bars are 1σ , and the downward pointing arrows denote upper limits. The tick marks on the horizontal axis are arbitrary. The data point at the far left is the northernmost point for which we have data. Then we proceed beam by beam south to the C position before changing directions and continuing beam by beam toward the west.

cm^{-3} from statistical equilibrium calculations with CS. Our results are consistent with the above results. We note that on average the density of the portion of L134N which we have observed is about 3 times lower than the average density of the observed positions in TMC-1 (Pratap et al. 1997). The effect of density uncertainties on abundances, including the results of multiplying the CS densities by 2.5, is discussed below in § 4.

4. ABUNDANCES

4.1. Column Densities

Having determined densities and kinetic temperatures in the cloud, we now use statistical equilibrium calculations to find the column densities of the molecular species observed. The calculations described in § 3 provide us with column densities for HC_3N , N_2H^+ , and CS. Since the lines of the main isotopomers may be very optically thick, we derive column densities for H^{13}CO^+ , H^{13}CN , HN^{13}C , and C^{18}O and scale by the isotopic ratios $^{12}\text{C}/^{13}\text{C} = 64$ (consistent with values measured by Langer & Penzias 1993) and $^{16}\text{O}/^{18}\text{O} = 500$ (Wilson & Rood 1994).

We take collisional coefficients and dipole moments, which we use to calculate radiative rates, from the literature (cf. Table 3). For the species which have no published collisional rates, we use collisional cross sections computed for similar molecules (e.g., HNC uses HCN collisional rates and CN uses those for CO). For those species which have collisional coefficients computed for collisions with He, we multiply these coefficients by $2^{1/2}$ to correct for the velocity difference between H_2 and He.

We take the kinetic temperature to be 10 K (cf. § 3.3) and the density to be that found in § 3.4 and summarized in Table 4. We calculate column densities only for the 50 positions for which we have determined densities. Due to the weak emission in many species, we have spatially averaged spectra over a $1/5$ radius to increase the signal-to-noise ratio. Line widths are taken from the Gaussian fits to the line emission centered about $V_{\text{LSR}} = 2.5 \text{ km s}^{-1}$. Recall from § 3.2 that some species, such as CS and SO, show evidence for multiple velocity components. We fit multiple Gaussian components to these spectra by fixing the line width of the 2.5 km s^{-1} component to be that found for the optically thinner isotopomer. Since we know the density for only the 2.5 km s^{-1} velocity component, we do not include the other emission features in our column density determination. The only remaining free parameter to our model is the column density, which is constrained by the observed line intensity.

For C^{18}O , H^{13}CO^+ , SO_2 , HN^{13}C , and CH_3OH , we vary the column density to find the best fit to the single observed line intensity. For CH_3OH , we only have collisional cross sections for the A species, so we model the $J_K = 2_0 \rightarrow 1_0 \text{ A}^+$ transition and multiply the resulting column density by 2 under the assumption that the A/E abundance ratio is 1. The uncertainty in the column density for these molecules is found by varying the column density until the model matches the upper and lower bounds to the uncertainty in the observed peak intensity. For those cases where there was no detection, we fit a Gaussian to the spectrum with V_{LSR} and line width fixed at the average values of the

detected lines of sight for that molecule to establish 1 σ upper limits to the intensity.

For CN, C₂H, and H¹³CN, we have additional information from the resolved hyperfine components. Therefore, we fit the individual hyperfine components with Gaussian line profiles, each with the same line width. Our model assumes that the excitation temperature is the same for each hyperfine transition and that the levels are populated according to their statistical weights. For SO, we have three transitions and a transition of ³⁴SO, all of which are used to

constrain the column density. Since we have more than one observed quantity for the cases of SO, CN, C₂H, and H¹³CN, the column density is found by minimizing χ^2 , and the uncertainties are determined by finding at what values of the column density the reduced χ^2 changes by 1 (cf. § 3.4). The results of this analysis are summarized in Tables 5 and 6 for each line of sight.

To determine the total para-NH₃ column density, we assume that the higher energy metastable states $J \geq 3$ are unpopulated at kinetic temperatures of 10 K. The non-

TABLE 5
MOLECULAR COLUMN DENSITIES OF CARBON AND SULFUR SPECIES IN L134N

Offset α (arcmin)	Offset δ (arcmin)	H ¹³ CO ⁺	C ¹⁸ O	CS	CH ₃ OH	C ₂ H	SO	SO ₂
0.63	0.04	12.66(0.06)	15.40(0.03)	13.52(0.02)	13.93(0.04)	< 13.71	14.44(0.04)	14.06(0.11)
0.63	0.88	12.71(0.05)	15.30(0.04)	13.16(0.04)	13.76(0.05)	< 13.61	14.20(0.04)	13.96(0.10)
0.63	1.69	12.58(0.06)	15.32(0.02)	13.20(0.03)	13.88(0.04)	13.64(0.09)	14.32(0.03)	13.72(0.13)
0.63	2.53	12.44(0.07)	15.19(0.03)	13.18(0.05)	13.77(0.04)	13.59(0.09)	13.88(0.04)	< 13.47
0.63	3.33	12.22(0.06)	15.27(0.03)	13.15(0.03)	13.53(0.06)	13.88(0.08)	13.62(0.05)	< 13.16
0.63	4.17	12.13(0.07)	15.19(0.03)	13.23(0.04)	13.34(0.08)	13.73(0.11)	13.51(0.06)	< 13.26
0.63	-0.85	12.45(0.06)	15.32(0.03)	13.43(0.03)	13.88(0.05)	< 13.78	14.18(0.03)	13.76(0.11)
0.63	-1.69	12.08(0.08)	15.26(0.03)	12.92(0.03)	13.62(0.07)	< 13.58	13.69(0.03)	13.29(0.15)
0.63	-2.50	12.08(0.11)	15.22(0.04)	12.67(0.08)	13.49(0.06)	13.57(0.10)	13.91(0.05)	13.37(0.15)
0.63	-4.98	12.37(0.08)	15.17(0.05)	13.08(0.06)	13.50(0.07)	< 13.35	13.86(0.05)	< 13.33
1.53	0.04	12.64(0.06)	15.40(0.03)	13.65(0.04)	13.93(0.04)	13.72(0.10)	14.45(0.04)	14.04(0.12)
1.53	0.88	11.98(0.06)	15.50(0.03)	12.49(0.06)	13.98(0.04)	13.58(0.11)	13.56(0.03)	13.28(0.13)
1.53	1.69	12.05(0.05)	15.37(0.03)	13.00(0.05)	13.88(0.04)	< 13.69	13.66(0.03)	< 13.07
1.53	2.53	12.45(0.07)	15.33(0.03)	13.69(0.02)	13.65(0.05)	13.74(0.09)	13.93(0.05)	< 13.21
1.53	3.33	12.14(0.08)	15.24(0.03)	13.22(0.05)	13.51(0.07)	13.74(0.09)	13.31(0.10)	13.53(0.16)
1.53	4.17	12.02(0.09)	15.15(0.04)	13.15(0.04)	13.46(0.07)	13.73(0.11)	13.06(0.10)	< 13.50
1.53	-0.85	12.18(0.07)	15.42(0.03)	13.22(0.08)	14.05(0.04)	< 13.68	13.79(0.04)	13.68(0.12)
1.53	-1.69	12.84(0.06)	15.26(0.03)	13.67(0.07)	13.70(0.05)	13.80(0.10)	14.30(0.07)	14.08(0.15)
1.53	-2.50	12.32(0.06)	15.30(0.03)	13.14(0.03)	13.58(0.06)	< 13.44	13.74(0.05)	< 13.22
1.53	-3.33	12.21(0.07)	15.22(0.04)	12.82(0.06)	13.72(0.05)	< 13.24	13.66(0.05)	13.66(0.10)
1.53	-4.14	12.51(0.06)	15.12(0.04)	13.12(0.02)	13.58(0.07)	< 13.48	14.18(0.04)	13.71(0.12)
1.53	-4.98	12.60(0.07)	15.11(0.04)	13.19(0.13)	13.60(0.05)	13.73(0.11)	14.22(0.05)	< 13.70
2.31	2.56	11.65(0.10)	15.28(0.03)	12.96(0.07)	13.64(0.06)	13.64(0.12)	12.92(0.05)	< 13.31
2.31	3.46	12.08(0.14)	15.20(0.03)	13.44(0.05)	13.22(0.10)	13.86(0.09)	13.46(0.08)	< 13.48
2.31	4.31	12.23(0.14)	15.12(0.03)	13.96(0.08)	< 13.08	13.76(0.11)	13.79(0.06)	< 14.49
2.31	-0.01	12.41(0.10)	15.37(0.03)	13.60(0.03)	13.74(0.05)	13.72(0.12)	14.39(0.03)	< 13.91
2.31	-1.56	12.48(0.11)	15.39(0.03)	13.64(0.03)	13.53(0.07)	13.77(0.09)	14.29(0.04)	< 13.37
2.31	-2.46	11.69(0.08)	15.31(0.04)	12.38(0.12)	13.61(0.06)	< 13.60	13.33(0.04)	< 12.84
2.31	-3.30	12.15(0.07)	15.17(0.03)	12.69(0.09)	13.52(0.07)	13.80(0.11)	13.66(0.04)	< 13.25
2.31	-4.20	11.87(0.09)	15.09(0.04)	12.88(0.05)	13.74(0.05)	< 13.45	13.70(0.04)	13.20(0.14)
3.17	0.04	< 12.32	15.42(0.02)	13.39(0.02)	13.26(0.10)	< 13.57	14.08(0.04)	< 13.36
3.17	-2.50	12.39(0.16)	15.40(0.04)	13.25(0.07)	13.45(0.06)	13.68(0.10)	14.21(0.04)	< 13.67
-0.14	0.82	12.35(0.05)	15.39(0.02)	13.05(0.04)	13.73(0.05)	13.52(0.11)	13.88(0.03)	13.68(0.12)
-0.14	1.72	12.15(0.06)	15.44(0.03)	12.87(0.04)	13.90(0.05)	13.56(0.13)	13.68(0.03)	13.30(0.17)
-0.14	2.56	12.63(0.06)	15.40(0.03)	13.29(0.04)	13.70(0.05)	13.75(0.10)	14.21(0.04)	< 13.81
-0.14	3.46	12.18(0.08)	15.31(0.03)	13.14(0.03)	13.61(0.06)	13.63(0.10)	13.56(0.04)	13.45(0.14)
-0.14	4.31	11.87(0.08)	15.15(0.03)	12.89(0.04)	13.32(0.14)	13.58(0.10)	12.92(0.08)	< 13.20
-0.14	-0.02	12.11(0.05)	15.48(0.02)	12.87(0.05)	13.81(0.05)	13.55(0.10)	13.85(0.03)	13.33(0.17)
-0.14	-0.72	12.17(0.07)	15.32(0.04)	13.18(0.03)	13.87(0.04)	13.78(0.10)	14.06(0.04)	< 13.46
-0.14	-1.56	11.82(0.06)	15.35(0.04)	12.85(0.07)	13.66(0.07)	13.52(0.12)	13.57(0.03)	< 13.34
-1.01	0.04	12.66(0.06)	15.47(0.03)	13.37(0.03)	13.82(0.04)	13.60(0.09)	14.61(0.03)	14.36(0.07)
-1.01	0.88	12.67(0.06)	15.28(0.03)	13.60(0.03)	13.74(0.05)	13.74(0.08)	14.52(0.03)	14.20(0.11)
-1.01	2.53	12.24(0.06)	15.29(0.03)	13.26(0.03)	13.86(0.05)	13.59(0.10)	13.90(0.03)	< 13.42
-1.01	3.33	11.96(0.08)	15.30(0.03)	13.09(0.06)	13.52(0.07)	< 13.40	13.46(0.05)	< 13.56
-1.01	4.17	11.66(0.09)	15.19(0.03)	12.63(0.04)	13.23(0.10)	13.67(0.10)	12.86(0.07)	< 13.10
-1.01	-3.33	12.45(0.12)	15.25(0.04)	13.19(0.06)	13.31(0.10)	< 13.46	14.35(0.04)	< 13.48
-1.88	1.73	12.17(0.16)	15.32(0.03)	13.74(0.02)	13.61(0.06)	13.51(0.10)	14.43(0.03)	< 13.92
-1.88	2.56	12.37(0.14)	15.16(0.03)	13.66(0.02)	13.39(0.07)	13.65(0.12)	14.24(0.03)	< 13.95
-1.88	-0.01	12.77(0.08)	15.44(0.03)	13.32(0.03)	13.85(0.04)	< 13.74	14.70(0.03)	14.56(0.06)
-2.66	0.04	12.36(0.15)	15.38(0.03)	13.69(0.03)	13.71(0.06)	13.59(0.13)	14.36(0.04)	13.96(0.15)

NOTES.—Offsets are relative to $\alpha(1950) = 15^{\text{h}}51^{\text{m}}30^{\text{s}}$, $\delta(1950) = -2^{\circ}43'31''$. The values given are the log of the column density. The numbers for CH₃OH are twice the column densities obtained for the A species. Uncertainties are 1 σ and are included in parenthesis. Upper limits are 3 σ .

TABLE 6
MOLECULAR COLUMN DENSITIES OF NITROGEN-BEARING SPECIES IN L134N

Offset α (arcmin)	Offset δ (arcmin)	N ₂ H ⁺	CN	HC ₃ N	H ¹³ CN	HN ¹³ C	NH ₃
0.63	0.04	13.13(0.02)	< 13.18	13.22(0.08)	< 12.62	13.33(0.07)	15.03(0.02)
0.63	0.88	13.03(0.02)	12.91(0.09)	13.20(0.09)	12.71(0.12)	13.10(0.10)	14.99(0.02)
0.63	1.69	12.87(0.02)	< 13.04	13.14(0.07)	< 12.75	13.33(0.09)	14.98(0.04)
0.63	2.53	12.89(0.02)	13.11(0.09)	13.06(0.06)	12.66(0.11)	13.13(0.10)	14.92(0.03)
0.63	3.33	12.84(0.02)	12.99(0.11)	13.11(0.01)	< 12.47	12.91(0.10)	15.02(0.02)
0.63	4.17	12.66(0.02)	13.12(0.09)	13.16(0.01)	< 12.39	12.78(0.15)	15.00(0.08)
0.63	−0.85	12.93(0.03)	< 13.22	13.02(0.06)	< 12.51	13.19(0.09)	14.93(0.04)
0.63	−1.69	12.63(0.02)	13.07(0.09)	12.58(0.08)	12.31(0.12)	12.57(0.16)	14.93(0.06)
0.63	−2.50	12.53(0.02)	13.03(0.09)	12.68(0.12)	< 12.27	< 12.79	14.93(0.06)
0.63	−4.98	12.43(0.03)	12.73	12.68(0.13)	< 12.17	< 12.30	14.93(0.06)
1.53	0.04	12.67(0.03)	13.37(0.09)	13.24(0.09)	< 12.61	13.13(0.13)	14.93(0.06)
1.53	0.88	12.42(0.03)	12.92(0.10)	12.38(0.05)	< 11.76	12.34(0.14)	14.88(0.06)
1.53	1.69	12.46(0.03)	< 12.91	12.62(0.04)	< 12.16	12.34(0.12)	14.71(0.07)
1.53	2.53	12.79(0.02)	13.29(0.09)	13.28(0.05)	< 12.57	13.21(0.08)	14.82(0.05)
1.53	3.33	12.66(0.02)	13.19(0.09)	12.89(0.01)	< 12.48	13.01(0.09)	...
1.53	4.17	12.54(0.02)	13.11(0.10)	12.96(0.01)	< 12.34	12.65(0.14)	...
1.53	−0.85	12.60(0.02)	< 12.94	12.52(0.04)	< 12.48	12.75(0.12)	14.95(0.04)
1.53	−1.69	12.97(0.03)	13.54(0.11)	12.74(0.03)	< 13.02	13.47(0.12)	14.78(0.04)
1.53	−2.50	12.69(0.02)	13.12	12.74(0.12)	12.42(0.11)	12.73(0.11)	14.70(0.03)
1.53	−3.33	12.61(0.02)	< 13.02	12.67(0.10)	< 12.17	12.52(0.11)	14.83(0.05)
1.53	−4.14	12.66(0.02)	13.18(0.08)	12.85(0.17)	< 12.39	< 12.56	...
1.53	−4.98	12.57(0.03)	< 13.29	13.08(0.17)	< 12.39	< 12.39	...
2.31	2.56	12.20(0.03)	< 12.89	12.56(0.04)	< 12.04	< 12. 21	...
2.31	3.46	12.54(0.03)	< 13.06	12.58(0.04)	< 12.56	< 13. 44	...
2.31	4.31	12.60(0.03)	13.45(0.11)	12.56(0.05)	< 12.70	< 12.13	...
2.31	−0.01	12.35(0.04)	< 13.12	13.11(0.09)	< 12.69	< 12.65	14.73(0.10)
2.31	−1.56	12.58(0.03)	< 13.12	13.11(0.14)	< 12.41	< 12.72	...
2.31	−2.46	12.41(0.02)	< 12.59	12.29(0.12)	< 11.70	12.17(0.12)	...
2.31	−3.30	12.54(0.02)	< 12.83	12.57(0.12)	< 12.55	12.46(0.15)	...
2.31	−4.20	12.49(0.02)	12.93(0.10)	12.34(0.12)	< 12.15	< 12.19	...
3.17	0.04	< 12.26	< 13.26	13.26(0.15)	< 12.55	< 12.66	< 14.45
3.17	−2.50	12.39(0.04)	13.41(0.11)	13.22(0.10)	< 11.99	12.92(0.16)	...
−0.14	0.82	12.76(0.02)	13.07(0.11)	12.56(0.01)	12.40(0.12)	12.66(0.14)	14.94(0.04)
−0.14	1.72	12.58(0.02)	< 12.30	12.55(0.03)	< 12.10	< 12.05	14.73(0.07)
−0.14	2.56	12.83(0.02)	< 13.05	12.61(0.03)	< 12.68	< 12.98	14.80(0.06)
−0.14	3.46	12.67(0.03)	13.13(0.08)	12.65(0.02)	12.50(0.09)	12.70(0.12)	14.67(0.08)
−0.14	4.31	12.26(0.04)	13.02(0.09)	12.58(0.04)	12.35(0.09)	12.44(0.14)	...
−0.14	−0.02	12.73(0.02)	13.05(0.08)	12.78(0.05)	< 12.46	12.77(0.10)	14.88(0.04)
−0.14	−0.72	12.66(0.02)	13.06(0.10)	13.06(0.05)	12.44(0.10)	12.59(0.14)	14.64(0.06)
−0.14	−1.56	12.22(0.03)	12.79(0.12)	12.55(0.04)	12.28(0.12)	< 12.31	14.76(0.10)
−1.01	0.04	12.85(0.02)	< 13.23	13.20(0.10)	< 12.55	< 13.12	14.61(0.05)
−1.01	0.88	12.69(0.03)	13.31(0.10)	13.09(0.11)	< 12.55	13.06(0.15)	14.79(0.09)
−1.01	2.53	12.49(0.03)	< 13.01	12.63(0.13)	< 11.40	< 12.43	...
−1.01	3.33	12.45(0.03)	12.98(0.11)	12.49(0.04)	< 12.16	< 12.18	...
−1.01	4.17	12.27(0.03)	< 12.75	12.49(0.04)	< 11.80	< 12.13	...
−1.01	−3.33	< 11.95	< 12.77	12.40(0.21)	< 12.30	< 12.15	...
−1.88	1.73	12.47(0.04)	13.40(0.09)	13.19(0.10)	< 12.66	< 12.41	...
−1.88	2.56	12.44(0.04)	13.11(0.12)	13.21(0.10)	< 12.15	< 12.64	< 14.61
−1.88	−0.01	12.79(0.02)	< 13.06	13.18(0.10)	< 12.66	< 12.53	14.55(0.10)
−2.66	0.04	12.52(0.03)	< 13.28	13.13(0.12)	< 12.22	< 12.70	...

NOTES.—Offsets are relative to $\alpha(1950) = 15^{\text{h}}51^{\text{m}}30^{\text{s}}$, $\delta(1950) = -2^{\circ}43'31''$. The values given are the log of the column density. The numbers for NH₃ assume an ortho/para ratio of 3. Uncertainties are 1 σ and are included in parenthesis. Upper limits are 3 σ .

metastable states decay too rapidly via infrared transitions to maintain any significant population. Therefore, we merely add the column density contributions from the (1, 1) and (2, 2) levels. We have already determined the optical depth of the (1, 1) transition in § 3.3, and this quantity is directly related to the column density in the (1, 1) line (Mangum, Wootten, & Mundy 1992). For those transitions that are optically thin, including all of the (2, 2) transitions, we use the expression in Bachiller et al. (1987), which relates the integrated intensity to the upper-level column density.

In our case, less than 3% of the total para-ammonia column density is contributed by the (2, 2) transition. To relate the para-NH₃ column density to the total ammonia column density, we need to make some assumption about the ortho/para ratio. The three identical spin-1/2 hydrogen nuclei result in the ortho species having 2 times the nuclear degeneracy of the para species. However, there are twice as many para as ortho states, such that if NH₃ molecules formed in thermal equilibrium at high temperature, we would expect an ortho/para ratio of unity (Townes &

Schawlow 1975). If there exists some mechanism for spin interconversion and equilibration of the ortho/para ratio at low temperatures, most of the molecules will be formed in the ground (ortho) state and exhibit an ortho/para ratio of ~ 3 at 10 K (Kuiper et al. 1995). Observationally, it is difficult to determine the ortho/para ratio for dark clouds since the (3, 3) level (ortho) is unpopulated at kinetic temperatures of 10 K. Hence, we make the same assumption as Swade (1989b) and Rydbeck et al. (1977) for L134N and Pratap et al. (1997) for TMC-1, namely that the ortho/para abundance ratio for ammonia is 3. The average NH_3 column density found in our study is $8 \times 10^{14} \text{ cm}^{-2}$, varying only by about a factor of 2 with position. Our value is similar to that found by Swade (1989b) of $2.6\text{--}9.0 \times 10^{14} \text{ cm}^{-2}$.

Just as is the case for TMC-1 (Pratap et al. 1997) our model results suggest that even some of the rarer isotopomer transitions are optically thick for the temperature and densities in L134N. For instance, the optical depths for H^{13}CO^+ , H^{13}CN , and HN^{13}C vary between 2 and 10 for those positions at which these species were detected. Even C^{34}S , C^{18}O , the strongest hyperfine components of C_2H and CN , and the weakest hyperfine component of N_2H^+ , have moderate optical depths of about 0.5 toward most lines of sight. A notable exception is the $\text{CH}_3\text{OH } 2_0 \rightarrow 1_0 A^+$ transition, which is optically thin toward every line of sight. Most transitions have excitation temperatures which are between 3 and 5 K, due in part to the large dipole moments for many of the species. C^{18}O and CH_3OH are thermalized at an excitation temperature of 10 K, and the $J = 4 \rightarrow 3$ transition of HC_3N has excitation temperatures between 7–9 K.

We want to compare our results with the previous study of L134N by Swade (1989a, 1989b). Although we do not observe exactly the same positions as Swade, we may compare with his data for positions which are closest to our own. In general, the column densities that we derive (cf. Tables 5 and 6) are larger, by a factor of 2–3 for C^{18}O , H^{13}CO^+ , HC_3N , and N_2H^+ and by nearly an order of magnitude for SO , SO_2 , H^{13}CN , and HN^{13}C , than those reported in Swade (1989b), despite similar observed line intensities (Swade 1989a). Therefore, the discrepancies in the column densities must be related to differences in either the excitation or optical depth. Swade assumed that the energy levels for a particular species were populated according to the Boltzmann relation at an excitation temperature of 5 K (except for CO which he assumed had an excitation temperature of 9 K), resulting in fractional populations of the upper levels of most transitions which are within 30% of that same quantity derived in our model. However, by assuming optically thin emission, Swade underestimated the column densities by an amount proportional to the optical depths found in our study.

The uncertainties quoted in Tables 5 and 6 are due solely to the rms errors in the measured line intensities. Other sources of error include calibration and errors from the derivation of the density and temperature. We investigated changes in the column density caused by a change in the density of a factor of 2. This is the typical 1σ uncertainty of our densities listed in Table 4. This resulted in percentage changes on the column densities of order of 40%–50% for CN , C_2H , and N_2H^+ . This number jumped to 70%–90% for the remaining molecules, with the exception of CO and CH_3OH , which changed by only 5%–10%. Both of these

molecules are thermalized and their excitation is changing very little at these densities. With the exception of CO, larger densities resulted in smaller column densities. We also examined the effect of increasing the CS-derived densities by a factor of 2.5, based on the discussions in § 3.4.4. For those lines of sight which included CS density determinations, the column densities decreased by 60%–70% for CN , C_2H , and N_2H^+ , decreased by 10%–15% for CH_3OH , increased by 10%–15% for CO, and decreased by 110%–150% for the remaining molecules. A change in the temperature by ± 2 K from the nominal value of 10 K produced changes in the column density on the order of 10%–20%. The largest variation was seen for HC_3N at 40%–50%. Except for CO, higher temperatures resulted in larger column densities. An important point to keep in mind is that a change in the temperature would require a corresponding change in the density in our model. When these two parameters are allowed to change in a consistent manner, the uncertainties quoted above decrease, and CO and CH_3OH are hardly changed at all. When all sources of uncertainty are considered, we feel that our column densities are good to within a factor of 2–3.

4.2. Relative Abundances

We follow the example of Pratap et al. (1997) in expressing molecular abundances relative to HCO^+ . The reason for adopting such a scale rather than comparing to CO was that for TMC-1, the C^{18}O emission did not appear to come from the same volume of gas as the emission seen in the other species. Using HCO^+ as the fiducial molecule has the advantage that it has similar excitation requirements as other species in the survey, plus it has a simple chemistry that is directly related to CO (Millar, Farquhar, & Willacy 1997). Therefore, like CO, its abundance varies only slowly with time after about 10^4 yr. By expressing our abundances relative to HCO^+ , we make the comparison with TMC-1 much easier. In Table 7, we list the abundances relative to HCO^+ for the N, C, S, and W positions, derived from spectra that were spatially averaged over a 1.5 radius around each position, along with a global average for TMC-1 (Pratap et al. 1997). We have determined, however, that in L134N the C^{18}O does not suffer from the same problem as in TMC-1 in that its line profile is fairly Gaussian and centered around $V_{\text{LSR}} = 2.5 \text{ km s}^{-1}$, similar to that for other molecules. Therefore, we derive abundances relative to CO as well. Since in general for quiescent clouds the CO/H_2 ratio is thought to be of order 10^{-4} (assuming $\text{C}^{18}\text{O}/\text{H}_2 \sim 1.7 \times 10^{-7}$ and $^{16}\text{O}/^{18}\text{O} = 500$; Frerking, Langer, & Wilson 1982), approximate fractional abundances with respect to H_2 can be obtained by multiplying the abundances relative to CO by this CO/H_2 ratio. In Table 8, we list the abundances derived in our model relative to H_2 for the N, C, S, and W positions in L134N.

For these relative abundances, we have investigated sources of uncertainty due to calibration and errors associated with the determination of density and temperature. Calibration errors are small since our data were taken with the same telescope over a narrow range of frequencies. In this way, our beam sizes and receiver response are similar. Additionally, uncertainties due to the determination of density and temperature will be smaller for the case of relative abundances, since most of the species behave in a similar manner with respect to changes in these quantities. Following the procedure outlined in § 4.1, we estimate that

TABLE 7
ABUNDANCES RELATIVE TO HCO^+ IN L134N

SPECIES	POSITION				TMC-1 AVERAGE	MODEL 2E5 yr
	N	C	S	W		
CO	7.1E3(0.6E3)	1.1E4(0.1E4)	1.2E4(0.2E4)	8.4E3(0.5E3)	7.8E3	1.4E4
CS	0.138(0.012)	0.124(0.006)	0.100(0.013)	0.069(0.004)	0.320	0.471
N_2H^+	0.058(0.005)	0.077(0.004)	0.060(0.007)	0.031(0.002)	0.013	0.067
NH_3	7.565(0.810)	7.635(0.722)	9.127(1.104)	4.284(0.621)	2.770	2.288
CH_3OH	0.311(0.028)	0.641(0.029)	0.579(0.072)	0.426(0.025)	0.099	0.189
C_2H	0.333(0.033)	0.288(0.021)	<0.224	0.171(0.023)	0.300	0.113
CN	0.069(0.010)	0.061(0.005)	<0.066	<0.045	0.070	0.917
HC_3N	0.073(0.006)	0.054(0.003)	0.034(0.006)	0.030(0.002)	0.150	0.071
HCN	0.968(0.123)	0.925(0.103)	<0.869	0.555(0.070)	0.490	0.420
HNC	3.963(0.386)	3.251(0.221)	1.954(0.321)	1.324(0.129)	1.680	0.677
SO	0.264(0.024)	0.719(0.032)	0.738(0.090)	0.716(0.041)	0.130	0.273
SO_2	<0.134	0.323(0.029)	0.283(0.045)	0.424(0.031)	...	0.052

NOTES.—The values are calculated from spectra averaged over 1/5 in radius around each position. Uncertainties are 1σ and are included in parenthesis. Upper limits are 3σ . Values for HCO^+ , HCN, and HNC are derived from observations of the ^{13}C isotopomer, scaling by $^{12}\text{C}/^{13}\text{C} = 64$; values for CO from C^{18}O , scaling by $^{16}\text{O}/^{18}\text{O} = 500$.

a change in the density by a factor of 2 results in a change in the abundance relative to HCO^+ of 2%–10% for the all species other than CO and CH_3OH , which were shown in § 4.1 to behave less similarly to HCO^+ . CO and CH_3OH changed by as much as 30%. Similarly, a change in the kinetic temperature by ± 2 K produced uncertainties on the order 2%–10%, with the largest change of 20% seen for HC_3N . Thus, our considerable analysis shows typically only a few percent uncertainty in the relative abundances, so we believe that abundance variations that are greater than a factor of 2 are physically real. We discuss such chemical variations in the next section.

5. DISCUSSION

5.1. Abundance Variations

Most of the observed variation in abundances relative to HCO^+ occurs on a line connecting our N and C positions, followed by a westward turn cutting through our W position. In Figure 8 we show log-plots of abundances relative to HCO^+ for this north-center-west cut, starting from our northernmost observed position (+0.63, +4.17), moving

southward through the N position to the C position, then westward through the W position and ending at (−1.88, −0.01). The N, C, and W positions are labeled on the x -axis, and the corresponding molecular hydrogen densities (*upper left panel*) and HCO^+ column densities (*upper right panel*) are plotted for comparison. Despite the many upper limits (*downward-facing arrows*), we see definite trends in the data. For instance, most of the carbon-bearing species exhibit a threefold decrease in their abundances as we move from the N to the W position. A similar trend is seen for the nitrogen-bearing species, N_2H^+ and NH_3 . On the other hand, the sulfur-bearing species, SO and SO_2 , increase in abundance from N to W. The location between the C and W positions does not seem to follow the abundance trends for most species. This position has an anomalous density which, if in error, could lead to the discrepant abundances. Although it is not included in the plot, the southern region has abundances similar to those seen in the center of the cloud core. One interesting feature to be noted is the flat distribution of CH_3OH .

The basic trend we see is that the carbon and nitrogen-bearing species have their peak abundances in the northern

TABLE 8
ABUNDANCES RELATIVE TO H_2 IN L134N

SPECIES	POSITION			
	N	C	S	W
HCO^+	1.19(0.10)E−8	7.92(0.35)E−9	6.87(0.83)E−9	1.01(0.06)E−8
CS	1.65(0.03)E−9	9.86(0.17)E−10	6.88(0.32)E−10	7.02(0.20)E−10
N_2H^+	6.84(0.20)E−10	6.13(0.18)E−10	4.11(0.14)E−10	3.10(0.09)E−10
NH_3	9.10(0.94)E−8	5.90(0.56)E−8	6.47(0.78)E−8	4.34(0.63)E−8
CH_3OH	3.71(0.12)E−9	5.07(0.10)E−9	3.98(0.14)E−9	4.31(0.10)E−9
C_2H	3.96(0.19)E−9	2.29(0.13)E−9	<1.54E−9	1.73(0.20)E−9
CN	8.18(0.92)E−10	4.81(0.37)E−10	<4.53E−10	<4.51E−10
HC_3N	8.67(0.17)E−10	4.29(0.20)E−10	4.38(0.26)E−10	3.07(0.16)E−10
HCN	1.16(0.10)E−8	7.32(0.75)E−9	<5.97E−9	5.61(0.62)E−9
HNC	4.72(0.21)E−8	2.58(0.14)E−8	1.34(0.14)E−8	1.34(0.10)E−8
SO	3.15(0.09)E−9	5.70(0.11)E−9	5.07(0.15)E−9	7.23(0.14)E−9
SO_2	<1.60E−9	2.56(0.20)E−9	1.95(0.19)E−9	4.29(0.20)E−9

NOTES.—The values are calculated from spectra averaged over 1/5 in radius around each position. Uncertainties are 1σ and are included in parenthesis. Upper limits are 3σ . Values for HCO^+ , HCN, HNC, and CO derived from observations of rarer isotopomers, as in Table 7.

region of the cloud core and fall off fairly smoothly as we move through the center and into the western region of the core, while the sulfur-bearing species show the opposite trend, i.e., they have their lowest abundances in the north and peak toward the west. In order to emphasize the opposing nature of these trends, we plot in Figure 9 the abundance ratios of $\text{HC}_3\text{N}/\text{SO}$, CS/SO , CN/SO , $\text{C}_2\text{H}/\text{SO}$, and $\text{HC}_3\text{N}/\text{SO}_2$, for the same north-center-west positions as in Figure 8. The significance of such plots becomes apparent when we note that the carbon-bearing species tend to be more abundant in the earliest stages of a core's evolution when the carbon has not yet been converted into CO and these species have not yet been frozen out onto the grain mantles. Hence, these carbon-bearing species have been called "early" type molecules. Similarly, pathways to the formation of the sulfur-bearing species involve reactions with longer timescales than those leading to the carbon-bearing species, such that the peak abundances of SO and SO_2 occur at later times in the evolution of the core, and they are called "late" type molecules. Thus, the variation we see in the ratio of "early"/"late" type molecules (e.g., $\text{HC}_3\text{N}/\text{SO}$) indicates that the N position is more abundant in "early" species, while the W position is more abundant in "late" species. It should be noted that the recent results of Ruffle et al. (1997) show that some species, such as the cyanopolyynes (e.g., HC_3N), exhibit secondary late-time abundance peaks, suggesting that the use of HC_3N as the sole indicator of early-time chemistry could be misleading.

5.2. Comparison with TMC-1

In addition to the relative abundances at the L134N positions reported in Table 7, we also list the global average

abundances for TMC-1 reported by Pratap et al. (1997). In Figure 10, we have plotted the ratio of the abundance in L134N at the C position to the global average TMC-1 abundance. When compared position-by-position, TMC-1 has higher abundances of carbon-bearing ("early" type) species than all L134N positions (N, C, S, and W). However, we see that L134N has a larger abundance of molecules like NH_3 , N_2H^+ , and SO, which are considered "late" type species or are relatively time independent. No values were reported for SO_2 in TMC-1. Hence, it looks like TMC-1 is at an earlier stage of evolution than the whole of the L134N core. This agrees with the general conclusions of earlier authors (e.g., Swade 1989a, 1989b; Swade & Schloerb 1992).

An exception to our view that L134N is at a later evolutionary stage than TMC-1 is the behavior of CH_3OH , which is considered an "early" type molecule, despite the fact that we find it to be more abundant by a factor of 6–7 in L134N than in TMC-1. Friberg et al. (1988) found CH_3OH to be similarly abundant in both sources. Those authors derived (as did we) that CH_3OH was optically thin and thermalized at 10 K in L134N, resulting in an abundance relative to H_2 of 2×10^{-9} . We derive a value of $\sim 6 \times 10^{-9}$ at the C position. Part of the difference between our results and those of Friberg et al. (1988) is related to our assumption that the E/A ratio is unity, while those authors derived $E/A = 0.69$. Correcting for this factor brings our agreement to within a factor of 2.

5.3. Chemical Model

One of the goals of this project was to compare our results with current time-dependent chemical models. As a

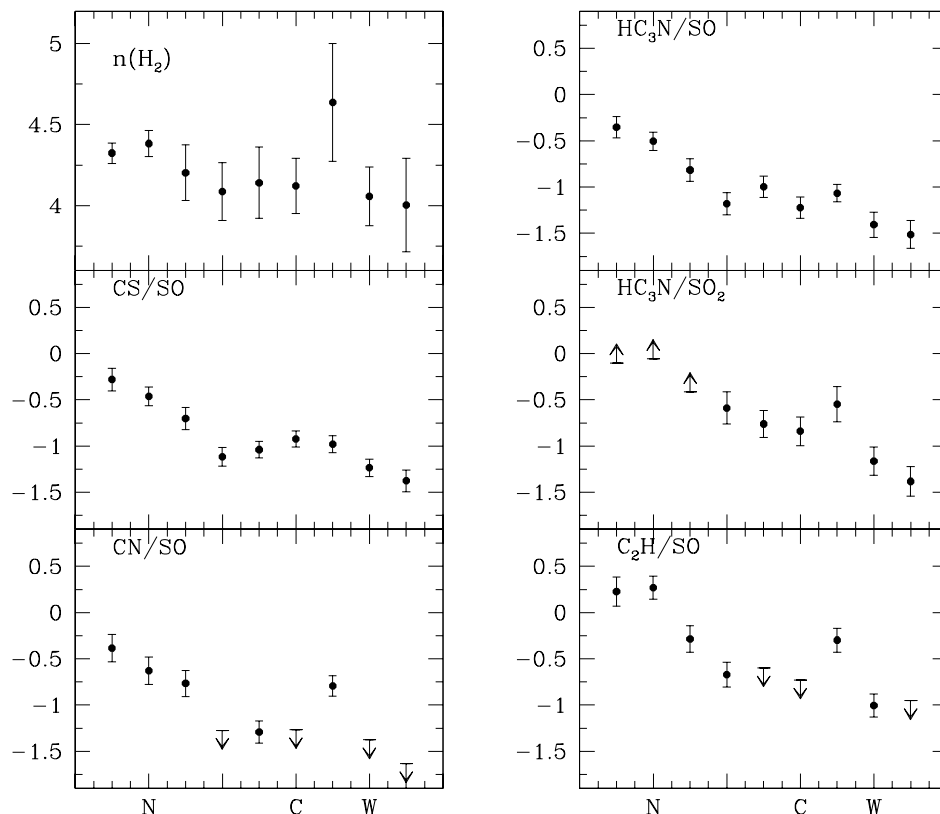


FIG. 9.—Ratios of log-abundances for "early"/"late" type molecules along the north-center-west cut in L134N. The horizontal scale is the same as presented in Fig. 8. Each panel lists the specific molecule ratio in the upper left corner. The upper left panel shows the log-density for reference. Upper limits are shown as arrows pointing down, and for $\text{HC}_3\text{N}/\text{SO}_2$, the upward arrows are lower limits. Uncertainties are 1σ .

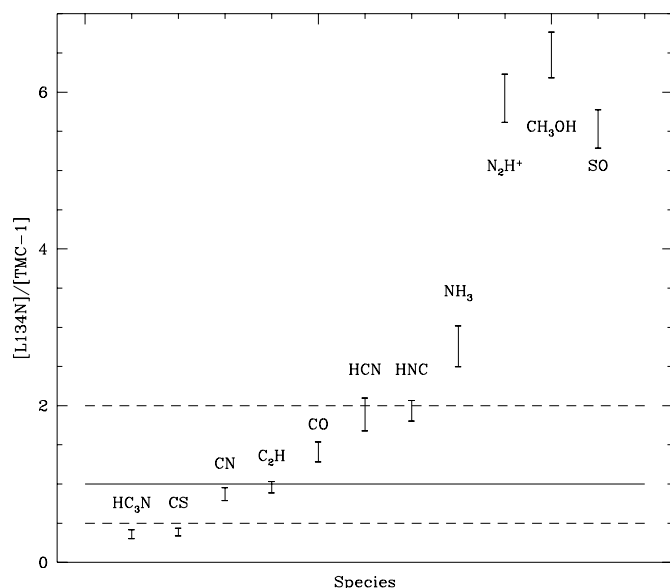


FIG. 10.—Plot of the abundance ratios L134N/TMC-1 for our observed molecules. The L134N value is taken from a $1/5$ radius around the C position, while the TMC-1 value is the global average taken from Pratap et al. (1997). The solid line denotes the 1:1 abundance ratio, and the dotted lines mark off values that are within a factor of 2 from this 1:1 ratio. No particular order of species is intended other than from lower to higher ratios. However, it can be noted that most “early” molecules (HC_3N – HNC) are clumped at lower ratios and “late” molecules (NH_3 – SO) are at higher ratios, suggesting that L134N is an “older” core; the “early” species CH_3OH is an exception. Uncertainties are 1σ .

first step in this process, we began with the chemical model presented for TMC-1 (cf. Pratap et al. 1997), which was adapted from the model described in Bergin, Langer, & Goldsmith (1995) ignoring effects related to grains. The chemical model was calculated for L134N using the average molecular hydrogen density and kinetic temperature derived in § 3 ($n_{\text{H}_2} = 2.1 \times 10^4 \text{ cm}^{-3}$ and $T_k = 10 \text{ K}$).

There has been considerable discussion very recently concerning the branching ratio for the principle reaction producing HCN and HNC, the dissociative electron recombination of HCNH^+ . There are no laboratory measurements, and the standard values used for interstellar chemistry assume that the products are formed in the ratio $\text{HCN}:\text{HNC}:\text{CN}$ equal to 0.25:0.25:0.5 (Herbst 1978). Recent quantum mechanical calculations (Shiba et al. 1998) and observations in the interstellar medium (Hirota et al. 1998) suggest that the branching ratios could be more like 0.6 to HNC, 0.4 to HCN, and none to CN. Irvine et al. (1998a, 1998b) found that their chemical model of the HNC/HCN ratio in Comet Hale-Bopp required similar branching ratios. In fact, from looking at Figure 14 in Pratap et al. (1997), we see that HCN and HNC are underproduced by the model while CN is overproduced. Therefore, in our model, we have adopted the branching ratios $\text{HCN}:\text{HNC}:\text{CN}$ equal to 0.4:0.6:0.0.

The results of the chemical model are shown in Figure 11, which plots the logarithm of the species abundance relative to HCO^+ as a function of time. Molecules such as CS, HC_3N , HCN, HNC, CN, C_2H , and CH_3OH are more abundant early in the chemical evolutionary stages of the cloud and are therefore called “early” species. SO and SO_2 do not begin to be produced in large amounts until $\gtrsim 10^5$ yr, so we call them “late” species. We have included hori-

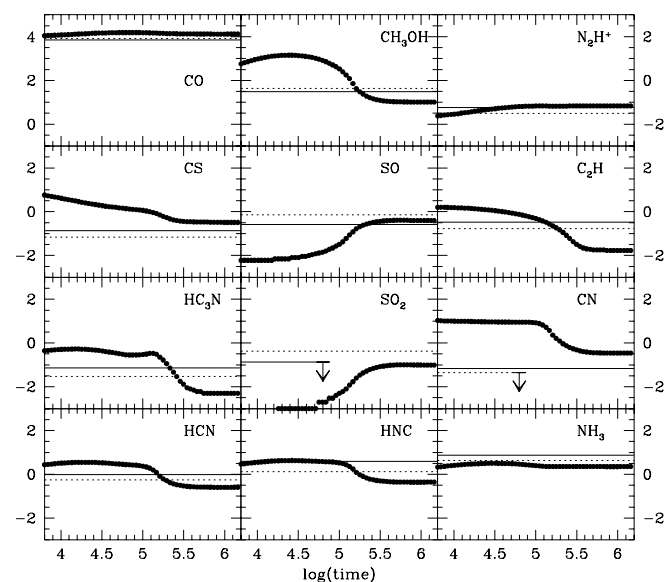


FIG. 11.—Chemical model described in § 5.3 is presented in log-space, showing abundance relative to HCO^+ vs. time, for the molecule listed in each panel. The y-axis scale is the same for every molecule, other than for CO, for which the scale is labeled. The solid line is the observed value at the N position, and the dotted line is the observed value at the W position, as presented in Table 7. Upper limits for SO_2 and CN are indicated with downward pointing arrows.

zontal lines which are representative of the abundances at the N (solid line) and W (dotted line) positions as they are listed in Table 7 (i.e., these are spatially smoothed averages over $1/5$ radius). This model does a fair job of reproducing the observed average abundances for a time near 2×10^5 yr, so we have included the model abundances relative to HCO^+ at this time in the last column of Table 7. Most species agree to within a factor of 2–3 for this time step in the model.

Since the carbon-rich “early” time molecules peak in the north and the oxygen-sulfur-rich “late” molecules peak in the south, we note that we can fine-tune our chemical model and get better agreement with the data by invoking a chemical age variation within the L134N core. At the temperature and density used in our model, an increase in the chemical age of $\sim 0.2 \times 10^5$ yr at 2×10^5 yr produces approximately a factor of 3 decrease in the carbon-bearing species and a factor of 3 increase in SO and SO_2 . The scale of these abundance changes is similar to the changes seen between the N and W positions for most species. Hence, we conclude that the northern region of the core is “younger” than the rest of the cloud with the western region being the most evolved.

Our observational results are also well matched by the more realistic evolutionary chemical models of Bergin & Langer (1997) and also with the models of Willacy & Millar (1998). These models include depletion of gas-phase species onto grain mantles and desorption mechanisms, and the Bergin & Langer models add a time-varying density. Both of these chemical models give similar results to the purely gas-phase model early in the evolution of the cloud up to about 10^5 yr, which is where our data fit best. As the cloud continues to evolve, the carbon and sulfur-bearing species continue to freeze out onto the grains and their gas-phase abundances drop. N_2H^+ and NH_3 are not depleted until the density reaches 10^6 cm^{-3} .

Our data are potentially suggestive of this same depletion pattern. If we look at the integrated intensity maps in the interval 2.4–2.6 km s⁻¹ in the species CS, SO, and CH₃OH overlaid with the N₂H⁺ contours (see Fig. 12), we see that for each peak in the N₂H⁺ emission, there is no corresponding peak in any of the other 3 species. This suggests that depletion is occurring on small angular scales. Higher angular resolution studies of a few key species, such as CS, SO, N₂H⁺, and CH₃OH are warranted to address these depletion cores.

6. CONCLUSIONS

We have mapped a 8.5 × 10' region in the core of the dark cloud L134N, including the dense peak in the north and the SO peak in the west, which were studied previously by Swade (1989a, 1989b) and Swade & Schloerb (1992). This study is an improvement on this previous work, since we have observed a larger number of species with a single telescope, resulting in similar beam sizes and antenna efficiencies. Statistical equilibrium calculations using multiple transition observations were performed to determine the kinetic temperature from NH₃ and the density from HC₃N, CS, and N₂H⁺. By relying on measurements of the rarer isotopomers, optical depth effects were minimized, allowing us to better determine whether the variations in the emission observed from one position to another within the core were related to changes in the abundance or excitation. We calculated column densities for 13 molecular species at 50 positions and present abundances relative to both HCO⁺ and H₂.

This study serves as a complement to that of TMC-1 by Pratap et al. (1997), such that we now have similar studies on two quiescent cores which are physically similar yet quite different in their chemical compositions. We used purely gas-phase chemical models to match observations among the two cloud cores and within the individual cloud cores. Overall, for the molecular species which we have observed in L134N, we find abundances that are within a factor of about 5 of those in the well studied cloud TMC-1. We find that TMC-1 is more carbon-rich than L134N, such that TMC-1 is “younger,” or less chemically evolved, than L134N.

Despite the lack of discernible variations in the kinetic temperature or density within the L134N core, we see evi-

dence for observed emission and abundance variations. Analysis of the abundances relative to HCO⁺ (cf. § 5.1) reveals significant variations from position to position in the L134N core. For instance, most “early” time chemical evolution species are most abundant in the north and fall off in abundance as we go further south and west. On the other hand, SO and SO₂, considered to be “late” time molecules, have their smallest abundances in the north and are most abundant toward the W position. This suggests that there is a chemical evolution difference, with the northern region of the cloud being “younger” than the west.

By invoking a difference in the chemical ages in these regions, our observed abundances are reasonably matched by our chemical model. For those molecules where our model cannot reproduce the observed abundances at the time matched by most other species, this large L134N data set will be instrumental in constraining reaction rates and production mechanisms leading to the formation of these species. In this context, we discussed further evidence which suggests that the “traditional” branching ratios assumed for the dissociative recombination of HCNH⁺, which produces HCN and HNC, should be modified (§ 5.3). Additionally, the models do not readily explain our observations of a rather flat distribution of CH₃OH and an abundance for this species that is considerably higher in L134N than in TMC-1. Perhaps the models are oversimplified in this regard.

There are several possibilities for the cause of the chemical age differences within the L134N molecular cloud core. Previous studies of L134N by Swade (1989a, 1989b) suggested that abundance differences might be caused by an increased gas phase C/O abundance ratio in the densest regions, related to the adsorption of water onto grains. In his model, the SO abundance peak toward the W position (called the “SO peak” by Swade) was indicative of a lower C/O ratio, such that oxygen-bearing molecules like SO and SO₂ were abundant. Such a C/O abundance gradient was suggested as one of the models to describe the abundance gradient observed along the TMC-1 ridge by Pratap et al. (1997), and those authors found that small changes in the C/O abundance ratio at a fixed time could produce large abundance variations. Density gradients have also been proposed to match the abundance gradients along the TMC-1 ridge (Hirahara et al. 1995; Pratap et al. 1997), and

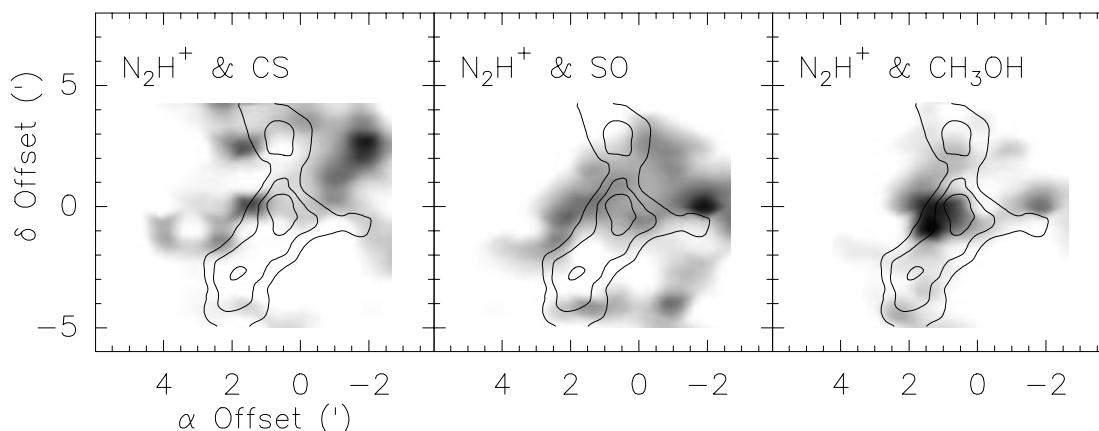


FIG. 12.—Integrated intensity maps in the interval 2.4–2.6 km s⁻¹ for CS (left panel), SO (middle panel), and CH₃OH (right panel) in gray scale. Each panel is overlaid with the N₂H⁺ contours.

again, small changes in the density in an evolving cloud result in large abundance variations at a fixed time. Although we do not see any clear evidence for such a variation in the density, there could be a factor of 2 uncertainty in the density toward some positions, such that variations in either the C/O abundance ratio or the molecular hydrogen density are both acceptable scenarios to match our observations.

We acknowledge support from NSF grant AST-9420159 and NASA grants NAGW-436, NAG5-3653, and NAG5-8718. We would like to thank the staff at FCRAO and Haystack for their help with the observations and Joe Santoro for helping with the NH_3 observations and analysis. We are very grateful to an anonymous referee for many helpful suggestions with the manuscript.

REFERENCES

- Bachiller, R., Guilloteau, S., & Kahane, C. 1987, *A&A*, 173, 324
 Bauer, A., Boucher, D., Burie, J., Demaison, J., & Dubrulle, A. 1979, *J. Phys. Chem. Ref. Data*, 8, 537
 Bergin, E. A., & Langer, W. D. 1997, *ApJ*, 486, 316
 Bergin, E. A., Langer, W. D., & Goldsmith, P. F. 1995, *ApJ*, 441, 222
 Bergin, E. A., Snell, R. L., & Goldsmith, P. F. 1996, *ApJ*, 460, 343
 Bergin, E. A., Ungerechts, H., Snell, R. L., Goldsmith, P. F., Irvine, W. M., & Schloerb, F. P. 1997, *ApJ*, 482, 267
 Bevington, P. R., & Robinson, D. K. 1992, *Data Reduction and Error Analysis for the Physical Sciences* (Second ed.; New York: McGraw-Hill)
 Botschiwina, P. 1989, in *Ion and Cluster Spectroscopy and Structure*, ed. J. P. Maier (New York: Elsevier), 59
 Caselli, P., Walmsley, C. M., Terzieva, R., & Herbst, E. 1998, *ApJ*, 499, 234
 Chin, N.-N., Henkel, C., Whiteoak, J. B., Langer, W., & Churchwell, E. B. 1996, *A&A*, 305, 906
 Danby, G., Flower, D. R., Valiron, P., Schilke, P., & Walmsley, C. M. 1988, *MNRAS*, 235, 229
 Flower, D. R., & Launay, J. M. 1985, *MNRAS*, 214, 271
 Frerking, M. A., Langer, W. D., & Wilson, R. W. 1982, *ApJ*, 262, 590
 Friberg, P., Madden, S. C., Hjalmarson, A., & Irvine, W. M. 1988, *A&A*, 195, 281
 Graedel, T. E., Langer, W. D., & Frerking, M. A. 1982, *ApJS*, 48, 321
 Green, S. 1975, *ApJ*, 201, 366
 ———. 1986, *ApJ*, 309, 331
 ———. 1994, *ApJ*, 434, 188
 ———. 1995, *ApJS*, 100, 213
 Green, S., & Chapman, S. 1978, *ApJS*, 37, 169
 Green, S., Montgomery, J. A. Jr., & Thaddeus, P. 1974, *ApJ*, 193, L89
 Green, S., & Thaddeus, P. 1974, *ApJ*, 191, 653
 Greve, A., Steppe, H., Graham, D., & Schalinski, C. J. 1994, *A&A*, 286, 654
 Herbst, E. 1978, *ApJ*, 222, 508
 Hirahara, Y., Masuda, A., Kawaguchi, K., Ohishi, M., Ishikawa, S., Yamamoto, S., Takano, S., & Kaifu, N. 1995, *PASJ*, 47, 845
 Hirota, T., Yamamoto, S., Mikami, H., & Ohishi, M. 1998, *ApJ*, 503, 717
 Irvine, W. M. 1992, in *Chemistry and Spectroscopy of Interstellar Molecules*, ed. D. K. Bohme, E. Herbst, N. Kaifu, S. Saito (Tokyo: Univ. Tokyo Press), 47
 Irvine, W. M., Bergin, E. A., Dickens, J. E., Jewitt, D., Lovell, A. J., Matthews, H. E., Schloerb, F. P., & Senay, M. 1998a, *Nature*, 393, 547
 Irvine, W. M., Dickens, J. E., Lovell, A. J., Schloerb, F. P., Senay, M., Bergin, E. A., Jewitt, D., & Matthews, H. E. 1998b, *Faraday Discuss.*, 109, 475, 511
 Kuiper, T. B. H., Langer, W. D., & Velusamy, T. 1996, *ApJ*, 468, 761
 Kuiper, T. B. H., Peters, W. L., Foster, J. R., Gardner, F. F., & Whiteoak, J. B. 1995, *ApJ*, 446, 692
 Kutner, M. L., & Ulich, B. L. 1981, *ApJ*, 250, 341
 Lafferty, W. J., & Lovas, F. J. 1978, *J. Phys. Chem. Ref. Data*, 7, 441
 Langer, W. D., & Penzias, A. A. 1993, *ApJ*, 408, 539
 Langer, W. D., Wilson, R. W., Henry, P. S., & Guélin, M. 1978, *ApJ*, 225, L139
 Lee, H.-H., Bettens, R. P. A., & Herbst, E. 1996, *A&AS*, 119, 111
 Lees, R. M. 1973, *J. Phys. Chem. Ref. Data*, 2, 205
 Lovas, F. J. 1985, *J. Phys. Chem. Ref. Data*, 14, 395
 Lovas, F. J., & Krupenie, P. H. 1974, *J. Phys. Chem. Ref. Data*, 3, 245
 Madden, S. C. 1990, Ph.D. dissertation, Univ. Massachusetts
 Maki, A. G. 1974, *J. Phys. Chem. Ref. Data*, 3, 221
 Mangum, J. G., Wootten, A., & Mundy, L. G. 1992, *ApJ*, 388, 467
 Menten, K. M., Walmsley, C. M., Henkel, C., & Wilson, T. L. 1988, *A&A*, 198, 253
 Millar, T. J. 1990, in *Molecular Astrophysics*, ed. T. W. Hartquist, (Cambridge: Cambridge Univ. Press), 115
 Millar, T. J., Farquhar, P. R. A., & Willacy, K. 1997, *A&AS*, 121, 139
 Ohishi, M., Irvine, W. M., & Kaifu, N. 1992, in *Astrochemistry of Cosmic Phenomena*, ed. P. D. Singh (Dordrecht: Kluwer), 171
 Pratap, P., Dickens, J. E., Snell, R. L., Miralles, M. P., Bergin, E. A., Irvine, W. M., & Schloerb, F. P. 1997, *ApJ*, 486, 862
 Ruffle, D. P., Hartquist, T. W., Taylor, S. D., & Williams, D. A. 1997, *MNRAS*, 291, 235
 Rydbeck, O. E. H., Irvine, W. M., Hjalmarson, Å., Rydbeck, G., Ellder, J., & Kollberg, E. 1980, *ApJ*, 235, L171
 Rydbeck, O. E. H., Sume, A., Hjalmarson, Å., Ellder, J., Rönnäng, B. O., & Kollberg, E. 1977, *ApJ*, 215, L35
 Schloerb, F. P., Friberg, P., Hjalmarson, Å., Höglund, B., & Irvine, W. M. 1983, *ApJ*, 264, 161
 Shiba, Y., Hirano, T., Nagashima, U., & Ishii, K. 1998, *J. Chem. Phys.*, 108, 698
 Snell, R. L. 1981, *ApJS*, 45, 121
 Snell, R. L., Langer, W. D., & Frerking, M. A. 1982, *ApJ*, 255, 149
 Swade, D. A. 1989a, *ApJS*, 71, 219
 ———. 1989b, *ApJ*, 345, 828
 Swade, D. A., & Schloerb, F. P. 1992, *ApJ*, 392, 543
 Terzieva, R., & Herbst, E. 1998, *ApJ*, 501, 207
 Thayer, M. R. 1996, Ph.D. dissertation, Univ. Chicago
 Thomson, R., & Dalby, F. W. 1968, *Canadian J. Phys.*, 46, 2815
 Tiemann, E. 1974, *J. Phys. Chem. Ref. Data*, 3, 259
 Townes, C. H., & Schawlow, A. L. 1975, *Microwave Spectroscopy* (New York: Dover)
 Tucker, K. D., Kutner, M. L., & Thaddeus, P. 1974, *ApJ*, 193, L115
 Ungerechts, H., Bergin, E. A., Goldsmith, P. F., Irvine, W. M., Schloerb, F. P., & Snell, R. L. 1997, *ApJ*, 482, 245
 Walmsley, C. M., & Ungerechts, H. 1983, *A&A*, 122, 164
 Willacy, K., & Millar, T. J. 1998, *MNRAS*, 298, 562
 Wilson, T. L., & Rood, R. T. 1994, *ARA&A*, 32, 191